

TargetSpace: Benchmarking Personal Intelligence by Target-Specific Forecasting Under Partial Observation

A prospective framework with own-routine baselines, permutation specificity, calibration, and evidence ablation

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Abstract

AI systems increasingly claim to know, remember, personalize to, and act on behalf of particular users, yet current evaluations cannot distinguish genuine person-specific predictive skill from retrieving stored facts, replaying preferences, imitating style, or predicting what usually happens. We call the missing capability **personal intelligence** — in a deliberately bounded sense, the capacity to form, update, and prospectively test a calibrated model of a specific person over time — and introduce **TargetSpace**, a prospective benchmark framework for it. The benchmark is motivated by an arriving capability class — AI systems that accumulate longitudinal observations about a target and convert them into future-relevant inference — made newly concrete by ambient AI recorders, wearable memory systems, and first-person audiovisual devices, whose longitudinal passive or semi-passive capture can transform ordinary conversations and activity traces into persistent models of future goals, commitments, and behavioural target states. In TargetSpace, a model receives observations of a target up to a sealed time and must emit calibrated probabilistic forecasts of the target’s future **target-state transitions** before those outcomes exist; forecasts are timestamped and hashed, outcomes resolve by deterministic rules, and scoring is strictly proper. A stack of controls makes target-specificity the measured quantity: a population-prior baseline (**R1**) the model must beat to exceed base rates; a person-specific **own-routine baseline (R2)** it must beat to exceed routine and memory replay; a **calibration gate**; and a **permutation specificity test**, under which forecasts scored against the wrong person’s outcomes must lose their skill. An **evidence-tier ablation** measures whether passive observation adds skill beyond digital exhaust, and an architecture-neutral grid compares language, vision-language, latent-predictive, symbolic, agentic, and human-self-report systems on identical sealed instances. Proper scoring, calibration, sealing, and evidence ablation are adopted from prior work and cited; the contribution is their conjunction, anchored on the own-routine baseline and the permutation specificity gate. This is a pre-pilot protocol proposal: no empirical results are reported, only the personal track is instantiated (synthetically), and a high score would certify calibrated predictive skill about a consenting individual — not licence to act on it.

1 Introduction

Modern AI systems are evaluated largely on what they know, retrieve, or solve in static tasks. Many consequential capabilities depend on something different: forecasting how a *particular* target — a person, a project, a patient, a market — will change over time. AI products increasingly stake exactly this claim: assistants that remember you, agents that act on your behalf, companions, coaches, and adaptive copilots all rest on the promise that the system has learned something durable about *this* individual and can anticipate what they will do or need. The promise is valuable, commercially central, and almost entirely unaudited. To make it auditable we introduce **TargetSpace**, a benchmark framework for **target-specific forecasting from passive longitudinal observation**. Given observations of a target up to a sealed time t , a model must produce calibrated probabilistic forecasts over the target’s future states *before those outcomes exist*; forecasts are timestamped and hashed, outcomes resolve later by deterministic rules, and scoring is strictly proper — contamination-resistant by construction rather than by curation. A model succeeds only if it exceeds both population base rates and the target’s own routine while passing calibration, evidence-ablation, and permutation-specificity checks. Advances on the benchmark would represent progress toward systems that reason over longitudinal context, detect when routines are about to change, and produce calibrated forecasts about specific real-world targets. This paper is a *benchmark proposal and protocol*: we introduce the benchmark protocol, define a pilot-ready personal-track instantiation with a synthetic worked example, and specify the evaluation design; empirical model comparisons are planned for the pilot, and no leaderboard results are reported.

The same structure recurs wherever a system claims to model a particular evolving entity, but the personal case is the most legible and the most consequential, and we take it as the flagship. The object there is what we call the **lived trajectory**: the unfolding, situated stream of attention, context, pressure, constraint, and emerging goals from which behaviour follows.¹ A person is not a profile to retrieve but a trajectory in motion, and TargetSpace asks whether a model can infer that trajectory from observable longitudinal traces well enough to forecast its next turn.

The difficulty is that many forecasts that look like understanding are not target-specific at all. A model can appear skilled by predicting what *usually* happens (most meetings occur, most emails are answered, most projects slip a little), and it can appear personalized by replaying the target’s *routine* (this person checks email at 9am, this project ships on Fridays). Both are useful; neither reveals whether the model has captured anything specific to *this* target beyond base rates and habit, and retrospective explanation cannot settle the question, since a model can rationalize what already happened without being held to what happens next. The personal claim in particular has many cheap substitutes — stored facts, retrieval over history, replayed preferences, stylistic imitation, short-horizon behavioural correlation — and we reserve **personal intelligence** for the stronger capability: forming, updating, and prospectively testing a calibrated model of a specific individual over time, with their changing priorities, latent goals, constraints, likely transitions, and uncertainty. Fluent generation does not close the gap either: a system that produces a plausible next action without predicting the state it leads to cannot be held to an outcome. TargetSpace evaluates the missing layer — not whether a system can *say* or *do* something plausible, but whether it can forecast what a specific target will *do or become* next, under specified evidence constraints.

The right question, then, is comparative and prospective: can the model forecast this target system better than (i) **the average system** — the population prior — and (ii) **the target’s own routine**? Only a model that beats both, stays calibrated, and loses its skill when its forecasts are

¹The contrast with third-person description loosely follows Nagel [92]; TargetSpace claims no access to consciousness, qualia, or private subjective states, and evaluates only forecasts of externally observable future states. Extended motivation is in Appendix F.

matched to the *wrong* target has demonstrated something specific to this system rather than to its crowd or its habit. **TargetSpace** operationalizes exactly this. It is a benchmark framework for prospective, sealed, proper-scored forecasting of a target system’s future target-state transitions under partial observation, with the controls that make target-specificity measurable: a population-prior baseline (**R1**), a strong instance-specific own-routine baseline (**R2**), a calibration gate, a permutation specificity gate, and an evidence-tier ablation. We name these the *target-specificity stack* (Section 3).

Concretely, TargetSpace is *not* primarily a video-prediction benchmark, a world-model benchmark, an agent benchmark, a robotics benchmark, a planning benchmark, or a static-LLM benchmark: it is a benchmark for **target-specific forecasting from passive longitudinal observation** — whether a model can infer evolving goals, constraints, context, and likely future target states from temporally extended, non-IID, multimodal observation of a particular target (Sections 2.4 and 6.3). Personal world modeling is the motivating first application, because understanding individual people is valuable and is currently underserved: today’s personal AI is trained largely on *digital exhaust* — calendars, messages, clicks — which often overrepresents already-externalized decisions and recurrent behaviour. But the formulation applies to any target system for which repeated observations and resolved outcomes make an own-routine baseline and a permutation test well-defined. We therefore describe TargetSpace as domain-general in formulation and instantiated first in personal world modeling, and we make no claim of cross-domain empirical validation in v1.

The closest empirical neighbours to this proposal are the experience-sampling and ecological-momentary-assessment tradition [61], personal-sensing work [62], and digital phenotyping [64], which likewise gather longitudinal, individual-level signals from everyday life. TargetSpace differs in what it places under evaluation: rather than predicting outcomes per se, it makes target-specificity itself the measured quantity, through the R1 population-prior and R2 own-routine baselines, proper scoring, calibration, sealed forecasts, and the permutation specificity test. The aim is complementary to these lines of work — we adopt their concern with situated, person-level data while shifting the object of measurement from prediction accuracy to whether a forecast is specific to a particular target rather than to a generic prior.

1.1 Motivation: ambient AI memory and the personal context moat

The capability class this benchmark evaluates is arriving — most visibly as commercial hardware — which is why the evaluation gap is now urgent rather than hypothetical. AI recorders began with episodic meeting and call transcription [98]; the frontier has shifted to longitudinal personal context. The clearest explicit statement is Omi’s manifesto [96], which frames *personal context as the moat* — “continuity: memory, goals, patterns” — and positions the device not as a recorder but as a system that accumulates a user’s context to build personal AI. Bee, acquired by Amazon in 2025, ships an ambient wristband that turns everyday conversations into summaries, reminders, and to-dos [97]; Limitless, whose pendant offered continuous conversational memory, was acquired by Meta in late 2025 [99]. A second wave adds vision: Ray-Ban Meta glasses put camera-and-microphone AI capture into mainstream distribution [100], Brilliant Labs’ Halo advertises an agent with cross-session “narrative memory” [101], and AR platforms are converging on the same first-person substrate [102], with egocentric-video research supplying the underlying datasets and methods [56,57]. These systems show the capability class arriving at consumer scale; they are evidence of the trend, not empirical validation of TargetSpace, and none of this is a claim that any vendor misuses data (Appendix L gives the analytical survey).

Wearable recorders and first-person audiovisual devices are not the scope of TargetSpace; they are an especially visible instance of a broader shift. The same benchmark problem arises whenever

an AI system accumulates longitudinal observations about a target — personal assistants and agents working over email, calendar, chat, documents, and browsing history; enterprise copilots and organizational memory; tutors over student interaction history; care systems over behavioural and physiological traces; robots observing environments and actions; recommender systems over attention and behaviour histories — and uses those traces to infer future goals, commitments, actions, needs, or state transitions. Ambient AI devices motivate the benchmark; they do not define its boundary.

The shift these systems share is the one this benchmark measures. A useful summary is that the system is not merely looking for the user’s *knowledge*; it is interested in the user’s *experience*. Knowledge is episodic and declarative; experience is longitudinal, situated, relational, embodied, multimodal — and predictive. Older privacy discussions asked whether a person was recorded; TargetSpace asks a different question: *what target states become inferable once many small observations are accumulated over time?* Ambient AI systems convert experience into prediction, and an evaluation standard for that conversion — what longitudinal capture actually buys, over base rates and over the target’s own routine — currently does not exist. That standard is what this paper specifies.

1.2 Understanding is the capability; forecasting is the measurement

*Understanding is a capability. Forecasting is a measurement.
The target is not a profile. The target is a lived trajectory.*

The capability we care about — an accurate, continuously updated grasp of how a specific system’s situation and dispositions evolve — is latent and impossible to score directly. Forecasting is its measurement, the role classification played for ImageNet [1] and standardized tasks for language and code [20,21]; across the sciences, predictive success is how a model is judged when the thing modeled cannot be inspected [14,15,18,22]. We do not claim prediction *is* understanding; we claim it is its strongest available operational signal, and only when shown to be about the *target* rather than the crowd. TargetSpace is therefore **forecast-first, not forecast-only**: a good score is a precondition for an explanatory claim, never proof of one. Of the four things a model might do — retrospectively describe a target, infer its current state, predict its next action, forecast its next target-state transition — the benchmark scores only the last.

1.3 The observation bottleneck, and possibility-space resolution

Why is this hard? The binding constraint is an **observation bottleneck**: we rarely see a system’s latent state, only a thin, biased residue — for a person, digital exhaust produced *after* they already decided what was worth recording. A model that learns only the surface regularities of this residue learns a routine, not a generator, which is exactly the failure target-specificity is meant to expose. What should a forecast be about? A partially observed system occupies a **possibility space** — a distribution over the states it might next move into — and an adaptive system acts to reach certain **target states** within it; understanding is operationalized as forecasting how that space **resolves** toward those states, especially the **transitions** between them, where routine breaks and target-specific skill is decisive. Two processes must not be conflated: **epistemic resolution** (the *observer’s* posterior concentrating as evidence accumulates) and **participatory target-state formation** (change in the *target itself*, as attention, action, feedback, and constraint alter which futures stay salient and reachable). The benchmark keeps them apart by scoring only against sealed external outcomes (Section 2.3); *resolve* is shorthand for a distribution concentrating, with no physical claim, and the extended treatment is in Appendix F.

Compressed artifacts versus the process behind them. The residue is also the wrong *kind* of trace. Current systems are trained and evaluated primarily on artifacts people intentionally produce — text, code, clicks, messages — compressed traces that record what a person chose to externalize, after intent was already summarized, and omit the observational stream from which goals, constraints, interruptions, and state transitions could be inferred as they happen. The hard missing layer for a human-centered world model is therefore not merely a better architecture but a longitudinal, passive-observation corpus of real activity aligned to goals, decisions, task transitions, and outcomes. TargetSpace specifies an evaluable prediction target over such observation; it does not claim the corpus exists, and Section 4 is explicit about what would have to be built.

Memory as a forward-looking belief, not an archive. Personal AI inherits a tempting default: treat memory as storage and success as faithful recall. Human memory is not built that way: autobiographical memory is reconstructed in the service of current goals rather than replayed verbatim [81], and the same constructive system simulates possible futures — memory is selective and prospective by design [82]. TargetSpace adopts this functional stance: it rewards a system not for reproducing logged history (the recall a retrieval system already supplies) but for converting passive longitudinal observation into a continually updated estimate of the person’s latent targets that improves prediction of future transitions. A benchmark scored on recall measures the archive; TargetSpace is scored on the forecast.

1.4 Attention as evidence and as hypothesis

Attention is limited and selective: it governs which information is sampled, learned, and turned into action [71,72], and in adaptive systems some target states *form* over time rather than waiting as fixed labels — repeated allocation can stabilize a provisional objective, withdrawal can displace one, and goals and attention are reciprocal [72,73]. In the main text we treat attention in exactly two roles: as *evidence* of the current target state, and as a *candidate source of predictive lift* — an empirical hypothesis tested by ablation, never assumed (Sections J.1 and 4.2). No physical, quantum, or consciousness claim is made; attention may proxy unobserved causes, so a causal role would require intervention or explicit structural assumptions. The benchmark contribution (a prospective, calibrated, target-specific forecasting test with R1/R2 and the permutation gate) is separate from this scientific motivation and from any philosophical interpretation, and stands even if the attention-formation hypothesis is false; the extended dynamical treatment is in Appendix F.

1.5 Why target-specificity matters

A base rate is the average trajectory; a routine is a profile of the *typical* trajectory; target-specificity is the demand that a model track *this* target’s trajectory precisely where it departs from both. **Generic prediction can masquerade as understanding.** Two cheap strategies produce convincing forecasts without any target-specific knowledge. The first is learning **base rates**: a model that knows the population statistics — how often meetings occur, emails get answered, projects slip, grasps succeed — will look calibrated and often correct, while knowing nothing about any individual target. The second is learning a **routine**: a model that replays a particular target’s recent regularities will look *personalized* while having captured only habit. A benchmark that scores raw accuracy or agreement cannot tell either strategy apart from genuine understanding.

Target-specific understanding is what remains after both are subtracted. Operationally it means three things together. First, the model must beat a strong own-routine baseline (R2): predicting a target’s habit is exactly what R2 already does, so skill over R2 is skill the routine does not contain. Second, the skill must be **calibrated**: a confident lucky streak is not understanding. Third, and

most distinctively, the skill must depend on the correct target-instance pairing: if we permute outcomes across targets — scoring the model’s forecasts for target i against the realized outcomes of a different target j — genuine target-specific skill should *collapse*. If it survives permutation, the model was modeling task or population structure, not the target.

One example makes the distinction tangible. Predicting that someone will eventually answer an email is base-rate prediction; predicting it because *this* person usually replies within a day is routine; the target-specific question is whether the model knew this person would ignore *this* message because their attention had shifted to another project this week. The same gap — generic versus routine versus target-specific — recurs for any tracked instance (a project’s slip beyond its organization’s normal delay pattern, a specimen’s response beyond its species base rate), and it is exactly what R2 and the permutation gate are built to expose. Cross-domain instantiations are proposed extension criteria, not validated results (Section 4.3).

1.6 Contributions

(1) Target-specific forecasting as a well-posed evaluation object. We distinguish three things often conflated — *generation* (predicting surface outputs), *imitation* (predicting plausible actions from demonstrations), and *target-state forecasting* (predicting the consequence, i.e. the latent target-state transition, of actions and events) — and define and score the third, under partial observation, distinguished from predicting population base rates, a target’s routine, next-action, or external events (Sections 1.5 and 2.1–2). **(2) The target-specificity stack.** A stack of controls centered on a strong instance-specific **own-routine baseline (R2)** and a **permutation specificity gate**, while adopting proper scoring, calibration, sealing, and evidence ablation from prior work (Section 3). **(3) An evidence-tier $\times$ architecture-class grid** for comparing LLMs, VLMs, JEPA-style models, multimodal agents, symbolic/probabilistic systems, hybrids, human self-report, and oracle bounds on identical sealed instances (Section 3). **(4) A first instantiation in personal world modeling**, where passive first-person evidence is tested against digital exhaust for target-specific predictive lift (Section 4). **(5) Cross-domain extension criteria** for other target systems, stated as formulation rather than as validated empirical breadth (Section 4.3). Throughout, the grid scores *consequence prediction* and target-state forecasting independently of whether a system is a reactive policy, a predictive world model, an explicit planner, or a hybrid (Sections 2.5 and K.1). The audited novelty is the *conjunction* assembled around one question, not any individual ingredient (Sections 3 and 6).

The distinctive contribution is therefore not a new scoring rule, architecture, or dataset but a prospective target-specificity test: each sealed forecast is evaluated for skill over R1 and R2, calibration, and collapse under target permutation, reported by evidence tier. Two of these controls are not already standard practice and anchor the design — the R2 own-routine baseline and the permutation specificity gate; the rest make the test interpretable, and it is the combination that distinguishes the proposal.

For reference, Table 1 states what this pre-pilot paper claims and does not claim.

2 The Benchmark Task

TargetSpace is not a model, a dataset, or an architecture. It is a benchmark framework — a task definition, a scoring methodology, standardized baselines, an evaluation grid, and integrity and governance protocols — organized as a shared apparatus with multiple application tracks (Section 4.3). Many datasets may instantiate it and many architectures may compete on it, the

Table 1: What this pre-pilot paper does and does not claim.

Item	Status
TargetSpace benchmark framework	Claimed (specified here)
Personal world modeling / personal intelligence	Specified as flagship track
Synthetic demonstration harness	Implemented; sanity check only
Human pilot results	Not claimed (no pilot run)
Cross-domain empirical validation	Not claimed
Passive observation improves forecasting	Hypothesis; to be tested by evidence ablation
Attention causes target formation	Not claimed; tested for incremental predictive value
Public raw first-person dataset	Staged, consent-governed release planned; not in initial release
Deployment legitimacy / permission to act on forecasts	Not claimed

sense in which ImageNet is distinct from any network trained on it and SWE-bench from any agent that solves its issues.

The systems under evaluation are what we call **ambient target-state systems**: systems that passively or semi-passively capture longitudinal observations about a person, group, or environment and transform them into a persistent representation capable of retrieving, summarizing, predicting, or steering future-relevant states. “Ambient” does not require continuous 24/7 recording; repeated passive or semi-passive capture across days to months qualifies. Observations may include audio, video, screen traces, location, documents, calendar events, interactions, or sensor data; target states include goals, commitments, intentions, needs, routines, decisions, and likely next actions or future behavioural states. The object of evaluation — and, as Section 7 argues, of governance — is not only the raw signal but the derived memory and model built from it. Table 2 makes the layering explicit: each layer up from the raw signal is more compressed, more searchable, and more predictive than the one below, and TargetSpace evaluates the top of this ladder.

Table 2: From capture to target-state inference. Each layer is more compressed, searchable, and actionable than the one below; TargetSpace evaluates (and the governance discussion of Section 7 concerns) the top layers, not only the raw signal.

Layer	Representation	Example capability	Primary risk
Signal	raw audio, video, screen, location, sensor traces	record or observe events	unauthorized or unnoticed capture
Text / perception	transcripts, OCR, visual labels, logs	make events searchable	searchable bystander speech and activity
Structure	speakers, entities, topics, tasks, commitments	organize experience into relations and obligations	relationship and obligation mapping
Memory	persistent longitudinal profile, embeddings, summaries, routines	retrieve and personalize over time	pattern-of-life inference
Target state	predicted goals, needs, actions, future commitments	anticipate (or steer) future behaviour	manipulation, surveillance, unwanted steering

2.1 From goals to target states

The object the stack scores is a target system’s target-state transitions. The intuitive version — forecast a system’s *goals* — smuggles in conscious intent, which excludes most adaptive systems. We therefore speak of **target states**: configurations a system behaves as if acting to reach and to restore when perturbed, whether or not any mind intends them — the minimal generalization that lets one apparatus span persons and non-persons.

Target states need not presuppose conscious, verbally reported goals. In the personal track the claim is operational: a target state is a *behaviourally resolved* configuration the person tends to reach, maintain, resume, or abandon — a commitment, priority, or task regime. Cross-scale biological examples — e.g. the target morphology a regenerating organism rebuilds toward [25] — motivate this broader, mind-agnostic vocabulary but are not required: the personal benchmark stands on behavioural forecasting alone, and nothing here is a metaphysical claim about tissues having goals. This use of target states is broader than conscious intent and is distinct from, though compatible with, reinforcement-learning formulations of goals and rewards. *Transitions* matter because the moments when what a system is acting to reach changes are where non-routine behavior originates and where a routine baseline fails — the regime in which target-specific skill is decisive. A transition is not a single event type: a target may newly emerge, stabilize, persist, compete with another, be displaced, abandoned, or resumed, or be forced by constraint or created by opportunity. TargetSpace treats this typology as a forecasting target with deterministic resolution rules (Table 9), which is what separates it from recognizing *which* fixed goal is currently active (Section I.3). The path a target traces through these transitions is what we call its **lived trajectory**: not a fixed profile of what it is, but the unfolding sequence of what it is oriented toward. Forecasting the target means anticipating that sequence — which state becomes salient next — from observable evidence; it is this, and no claim about inner experience, that the benchmark scores.

2.2 Definitions and the forecast unit

A **target-system instance** is a specific partially observed adaptive system tracked over time (a person, animal, robot, tissue, organization, project, software agent, market). Its **current abstract state** is the latent configuration generating its behaviour at time t ; a **latent target state** is a configuration it acts to reach or maintain; a **transition path** is the sequence of target states it moves through — the formal counterpart of the *lived trajectory* of Section 2.1, and the object a model is asked to forecast. A benchmark instance is the tuple $(i, E_{\leq t}, q, A, r)$: instance i ; evidence available up to time t ; a query q about a future state with discrete answer space A ; and resolution time $r > t$ with a deterministic **resolution rule**. The system outputs a distribution over A . The unit of analysis is the resolved forecast; but because forecasts are nested within instance and serially dependent, statistical inference is performed at the instance level (hierarchical models or instance-clustered resampling), and the number of independent *instances* — not the raw forecast count — bounds power for population-level claims. Forecast queries are issued by the organizer, not chosen by the entrant, so skill cannot be inflated by forecasting only the easy instances. Three assumptions are explicit: **A1** the future is partially predictable, bounded above [13,91]; **A2** latent target states are evaluated only through observable consequences; **A3** the instance changes, so the benchmark rewards adaptation. The measurement is agnostic to which latent structure produces the forecasts — the property that makes the grid of Section 3 architecture-neutral. A **forecast horizon** is part of the object: forecasts are specified at multiple horizons and abstraction levels — *short-horizon* concrete next states, *medium-horizon* routine deviations, and *long-horizon* abstract target-state transitions — reflecting that detailed prediction is reliable only at short horizons

while long-horizon prediction requires abstraction. A hierarchical representation connects abstract long-horizon forecasts to concrete short-horizon ones, and each level is scored independently rather than assuming one fixed representation is optimal across horizons (Table 8). The four task families (Section 5.5) form exactly this horizon–abstraction ladder, scored throughout against the routine baseline (R2) and the permutation specificity gate.

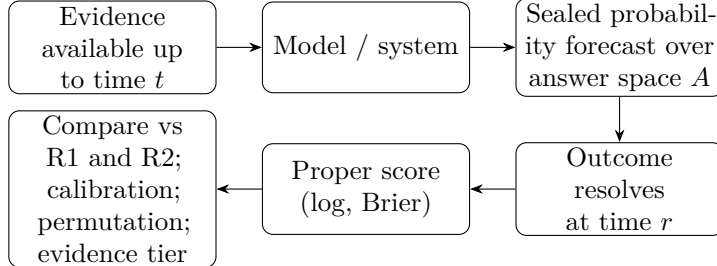


Figure 1: The TargetSpace evaluation loop. A model or system receives evidence available up to time t and emits a sealed (timestamped, SHA-256) probability forecast over the answer space A . After the outcome resolves at the resolution time r , the sealed forecast is graded with a proper score (log score in bits, Brier) and compared against the population-prior baseline (R1) and the own-routine baseline (R2), with the calibration gate, the permutation specificity gate, and the evidence-tier ablation. Only the sealed forecast and the resolved outcome are scored; the system’s internal explanation or representation is not.

The capability as belief-state inference. It is useful to name what a competent system must internally maintain, while being precise that the benchmark never scores it directly. From observations $O_{1:t}$, a system forms a **belief state** B_t — a distribution over latent variables (the target’s active and competing goals, binding constraints, salient episodes, recent transitions, and the system’s own uncertainty), updated as observations arrive and serving as a sufficient statistic of the history [88] — and emits the calibrated distribution over the target-relevant future that B_t implies. B_t is latent and architecture-specific, and TargetSpace does not score it: two systems with very different internal B_t remain comparable because both emit a distribution over the same answer space, and R2 and the permutation gate test whether whatever B_t a system maintains is genuinely about *this* target. A forecast that beats R2 and collapses under permutation makes B_t a credible candidate for later *explanatory audit* (Section 7) — a secondary analysis that never substitutes for prospective validation.

We keep several often-conflated notions distinct, because the benchmark scores some and not others. A **stated goal** is what a person declares they want; an **inferred goal** is what a model concludes they are pursuing; an **enacted priority** is what their allocation of a scarce resource (attention, time) actually reveals; a **predicted next action** is a surface behaviour; a **predicted future state** is a value in a defined answer space at a resolution time; and a **latent constraint** is a binding limit that shapes which transitions are possible. TargetSpace scores *predicted future states and their transitions* — with goal and constraint treated as latent structure evaluated only through those observable consequences (A2) — not stated goals, and not next-action mimicry. We use *goal-state transition* and the more neutral *meaningful transition* interchangeably where no theory of goals is required. Throughout, a **scored target** is a pre-registered discrete future state or transition with an externally observable **outcome** that resolves it; latent goals, constraints, and attention proxies are explanatory or evidential structure, scored only when mapped to such a target.

An **implicit target** is a latent orientation inferred from behaviour, repetition, avoidance, and attention patterns — a configuration the person behaves as if organizing around even when

undeclared, and possibly without introspective access to it [23]: unresolved concerns, avoidance patterns, preparation for a pending decision, emerging commitments, deferred goals. Like stated and inferred goals, it is *evidence-bearing latent structure*, never a scored target in its own right: “the person is avoiding this reply” is scored only as, e.g., $P(\text{no substantive response to message } m \text{ by } r)$.

Protocol elements. Table 3 fixes each element of a benchmark instance as a concrete definition with its pre-registered option(s). Where an element is a design choice, the option is set before sealing and reported with the result, so the protocol is concrete rather than left implicit.

Table 3: Protocol elements: each a concrete definition with its pre-registered option(s). Design choices are fixed before a forecast is sealed and disclosed with the result.

Element	Definition	Pre-registered option(s)
Target	the tracked instance i	person / agent / system / organization / process
Target state	latent configuration z_t the target acts to reach or maintain	scored only via observable consequences (A2)
Forecast	a distribution over the discrete answer space A at time t	proper-scored; nonzero probability floor
Observation window	evidence $E_{\leq t}$ available up to forecast time t	zero- / short- / longitudinal-history arms
Prediction horizon	$h = r - t$, from t to resolution time r	short / medium / long; scored per horizon
Label / outcome	the resolved value at r	set by a deterministic resolution rule
Ground truth	how the outcome is established	observed action, sensor threshold, or pre-registered rule; inter-rater reliability where contestable
Evaluation metric	how skill is scored	log score (bits), Brier; Skill over R1/R2
Baseline	reference the system must beat	R1 population prior; R2 own-routine
Uncertainty reporting	calibration and interval reporting	calibration gate; day-blocked / person-clustered intervals
Leakage control	preventing use of future information	sealing (SHA-256) + strict walk-forward; random CV prohibited

2.3 The walk-forward sealed protocol

A retrospective benchmark replays logged histories, so apparent forecasting becomes partly retrieval — the failure that motivated contamination-resistant designs [16,17]. TargetSpace privileges a prospective, sealed mode (Figure 1): forecasts are timestamped and sealed (SHA-256) before outcomes exist and resolved automatically. We enforce strict walk-forward (prequential) evaluation — only evidence timestamped $\leq t$, indices rebuilt per slice; random cross-validation is prohibited. Federation is a core initial deployment option, not a rejection of centralized datasets: for sensitive longitudinal first-person data the harness can run where the data lives and export only sealed forecasts, resolved outcomes, and aggregate reports, addressing the privacy and decentralization problems without requiring a central raw corpus. This staged approach separates the benchmark definition and scoring protocol from the later, consent-governed construction of a shared corpus (the dataset roadmap is Section 7). Two properties follow. First, target-specific forecasting is inherently longitudinal rather than IID: a target’s earlier history may become relevant to later forecasts, so the evaluation preserves chronology rather than randomizing examples away from their temporal context. Second, a forecast is scored against sealed external outcomes, never for internal self-consistency: a system that produces confident rollouts inside its own latent simulator earns

no credit until those rollouts resolve against what the target actually did, which guards against a model that is competent only inside its own simulator.

2.4 What TargetSpace is not

Not user modeling. A user model predicts a target’s *outputs* (clicks, ratings, responses) to tailor a system; a target-state model forecasts the *evolving latent state that generates those outputs*. **Not event-outcome forecasting.** Sealed proper-scored forecasting of external events is established [28,40]; TargetSpace scores transitions in the latent target state of a *tracked instance* with instance-specific controls (R2, permutation) those benchmarks do not use. **Not a chat, retrieval, task-completion, or desktop-automation benchmark.** Those evaluate what a system *does* when asked; TargetSpace evaluates whether a system can infer and predict a person’s goal states and their transitions from *passive observation up to a sealed time*, before any request is made. **Not a competitor to JEPA** or to latent-prediction research (Section 6): it evaluates such systems rather than replacing them.

2.5 Why target-state forecasting differs from generation and imitation

Four capabilities are easy to conflate, and the benchmark scores a specific one. **Generation** predicts surface outputs, judged by plausibility or fidelity; an **imitation or reactive policy** maps observations and instructions to actions; **consequence prediction** forecasts the future state an action, event, or transition produces; **planning** searches over predicted futures to choose actions. These may coexist in one system (a vision-language-action policy paired with a learned subgoal model [78]). TargetSpace directly scores *consequence prediction* and *target-state forecasting*: it does not infer planning ability from action success and does not require a human-readable internal simulator — any system that can emit a calibrated distribution over externally resolvable future states may participate, whether a wrapped reactive policy, a latent world model, an LLM, a symbolic planner, or a hybrid.

The part of the future that matters. TargetSpace does not ask a system to predict every ripple in the river — every pixel, token, or surface detail, much of which is irrelevant or intrinsically unpredictable. It asks whether the system identifies and scores the *target-relevant* transition — the same intuition that motivates predicting in a learned representation rather than reconstructing raw observations (Section I.2), lifted to the measurement layer. The benchmark’s object is thus a world-model question made architecture-neutral: given a target, partial evidence, and a horizon (optionally a hypothetical intervention), what distribution does the system assign to the target’s next state? A system that can only produce a plausible surface, or reconstruct a full but mostly-irrelevant future, has not answered it.

An optional action-conditioned mode makes consequence prediction explicit. A base instance $(i, E_{\leq t}, q, A, r)$ (Section 2.2) extends to $(i, E_{\leq t}, q, A, U, r)$, where U is a candidate action, event, or intervention context, and the system predicts $P(z_{t+h} \mid E_{\leq t}, U)$, the target-state distribution at horizon h under U . Four things stay distinct: passive forecasting (no U), observational counterfactual prediction (a hypothetical U scored on later realized cases), actual intervention, and causal-effect estimation. An action-conditioned forecast scored on observational data is *not* a causal estimate; causal identification requires intervention, natural experiments, or explicit structural assumptions (Section 7).

3 The Target-Specificity Stack

We present the controls not as a loose conjunction of desirable properties but as a **stack**: each layer removes a way a forecast can look like understanding without being target-specific. TargetSpace adopts most of these tools from prior work (Section 6); its contribution is assembling them around a single question — *is the forecast genuinely about the target system?* — and adding the two layers that directly test it.

Table 4: The target-specificity stack. Layers 1-3, 5, and 7 are established tools we adopt and cite; layers 4 and 6 — the own-routine baseline and the permutation specificity gate — are the anchors that make the stack a test of target-specificity rather than of generic predictive quality.

Layer	What it removes / certifies	Status
1. Prospective sealing	forecasts sealed and timestamped before outcomes exist — removes hindsight rationalization and contamination	adopted [2,28]
2. Proper scoring	log / Brier scoring of the whole distribution — rewards calibrated probabilities, not cherry-picked accuracy	adopted [14,15,18]
3. R1 population-prior baseline	skill must exceed population base rates — filters out generic base-rate prediction	adopted [93]
4. R2 own-routine baseline	skill must exceed the target’s <i>own</i> strong routine — filters out routine mimicry	anchor (this work)
5. Calibration gate	expected-calibration-error (ECE) threshold — filters out overconfident lucky prediction	adopted [43]
6. Permutation specificity gate	skill must collapse when forecasts are matched to the wrong target — tests dependence on the correct instance pairing	anchor (this work)
7. Evidence ablation	skill as a function of evidence tier — measures which streams add target-specific information	adopted [62,64]

The stack composes into a single decision rule (Figure 4): a forecast earns *target-specific* credit only if it beats R1, beats a strong R2, passes calibration, and *fails* under permutation. Removing any one layer reopens a way to score well without target-specific understanding, which is why we present them together rather than as separable contributions.

Holding the forecast unit and scoring fixed, the stack is applied across an evaluation grid that varies *what a system may observe* (the evidence tier, from low-content behavioural metadata, L0, to physiological sensors, L6) and *what kind of system it is* (the architecture class: LLM, VLM, multimodal agent, latent-predictive, symbolic/probabilistic, reactive policy, hybrid, plus human self-report and oracle anchors). Every cell is comparable because every system emits a distribution over the same answer space on the same sealed instance, through a disclosed output adapter reported as part of the system under test. The full architecture-class and evidence-ladder specifications, including the per-tier privacy-risk taxonomy, are in Appendix K; whether richer evidence helps is measured by the tier ablation (skill over R2 per tier), never assumed.

4 Personal-Track Instantiation

Personal world modeling is the instantiation we carry furthest, because understanding individual people is valuable and the observation bottleneck bites hardest there. Here the target is most vividly a lived trajectory, and the passive evidence tiers (Table 15) are its observable traces — attention, context, interruptions, pressure, and the timing of action — from which a model must infer the latent target states and forecast their transitions, never accessing the person’s experience

itself. The target-system instance is a consenting individual; the scored target states are categories such as commitment active versus abandoned, priority maintained versus displaced, and task continuation versus switch (Table 9), while attention, concerns, and avoidance are *evidence* about those states rather than states themselves (the evidence ladder is Table 15). The personal-AI question is sharpened by target-specificity: not merely ‘does passive capture help?’ but ‘does passive first-person evidence help a model forecast *this person’s* future target-state transitions beyond population priors and beyond this person’s own routine?’

Concretely, the **forecast targets** are observable future states and transitions — commitment follow-through, meeting and event realization, response behaviour, task continuation versus switch, priority maintenance versus displacement, and engagement versus avoidance of a defined obligation — each with a deterministic resolution rule (Table 9); attention allocation, social context, and emotional cues are **evidence**, and deadlines, dependencies, and competing obligations are **constraints**. The target-specific hypothesis is the same throughout: digital exhaust (L0–L1) often overrepresents this person’s routine, which a strong R2 already captures; passive evidence (L2–L4) may reveal *deviations* from routine — the shifted attention, the commitment about to slip — that are exactly where skill over R2 is earned. The hard case is accumulation: individually banal observations become predictive when combined across time, which is why the benchmark evaluates inference from longitudinal traces rather than question answering over transcripts. Audio-only traces are the early form of this evidence [96–99]; audiovisual first-person traces are the stronger future form because they add embodied context — what the person sees, where they are, which objects, documents, and screens are present, who is nearby, and what actions are physically taken [56,57,100–102]. The move from audio to audiovisual first-person capture expands the TargetSpace problem from conversational memory to embodied experience modeling; in the evidence ladder both are simply tiers (L2 versus L3–L4), and whether richer capture pays is measured by the ablation, never assumed.

The generation / imitation / forecasting distinction (Section 2.5) maps cleanly onto the personal case. The task is not to *summarize* a person (generation) or to predict *what people usually do* (a base rate, or the imitation of typical behaviour); it is to forecast *this person’s* deviations and next-state transitions from partial evidence, while beating that person’s own routine baseline.

4.1 A minimal runnable task

To fix ideas, one pilot-ready task — the *Recurring-Commitment Completion Forecast* — is runnable on current language models, agents, and retrieval or memory systems. Given timestamped evidence up to a sealed time T (calendar and task history, communication metadata, prior completion/defer/cancel history; optionally text traces and passive-observation summaries), the system forecasts how a scheduled recurring commitment resolves by resolution time r , over the answer space $A = \{\text{complete, defer, cancel, replace}\}$. R1 and R2 are fit walk-forward on evidence before T ; a deterministic rule over later observable evidence (calendar status, attendance, task completion, or a pre-registered behavioural marker) fixes the outcome; scoring is log score (bits) and Brier against the sealed outcome, reported as skill over R1 and over R2; and a permutation test across matched targets must reduce or collapse the skill. The full specification is Appendix G. This is the shape of every personal-track task: the transition types it draws from are catalogued in Appendix H.

4.2 Attention-conditioned predictive lift

In the personal track, attention allocation is the leading evidence stream: it reveals what currently competes for a person’s limited cognitive and behavioural resources [24], and its trajectory is a continuously observable, often leading indicator of target-state transitions — evidence of allocation,

not a read-out of goal identity. (In non-person domains the analogue is whatever scarce resource the instance allocates — compute for a software agent, managerial attention for an organization [74].) Whether attention is also a *process* that helps form the next target state is the hypothesis of Section 1.4, and we make it falsifiable rather than assume it. Writing the transition as $P(z_{t+1} \mid z_t, a_{\leq t}, e_{\leq t}, u_{\leq t}, c_{\leq t})$ — with a attention, e evidence, u action, c constraints — we compare a matched reduced model omitting the attention trajectory; **attention-conditioned predictive lift** is the difference in proper-scored, target-specific Skill, measured beyond R1, R2, stated goals, and contemporaneous behaviour. The protocol pre-registers the attention proxy, its granularity, missingness handling, separation from mere activity frequency, and leakage prevention; lift is scored only on sealed future outcomes. A positive lift demonstrates incremental predictive value, *not* causation — attention may proxy an unobserved cause, and causal identification would require intervention (Section 7). The active-inference account, in which action and attention shape expected future states, is one compatible framework [19,32,75], not a settled mechanism.

4.3 The multi-track apparatus

TargetSpace is not a single dataset or one domain-specific benchmark but a shared evaluation apparatus with multiple application tracks: **TargetSpace** is the framework, **personal intelligence** (TS-Personal) is the flagship capability and track, and the remaining tracks — health, energy, robotics, enterprise (Appendix K) — show the framework is not merely a personalized-assistant benchmark. Each track defines domain-specific targets, evidence sources, horizons, baselines, and outcome validators while preserving the common scoring spine of Section 3, the way a fixed metric spine is applied across scenarios in holistic LLM evaluation [43]. A track is admitted only when it requires a *distinct* validator, evidence band, and horizon profile **and** admits a strong R2 — otherwise it is a regime within an existing track, or not yet a track; the top level is capped to avoid dilution, and each track is tiered honestly by readiness.

Adding a track instantiates the spine, not changes it. We **deliberately exclude** scientific perturbation forecasting (e.g. cell-fate): it is one-shot and cross-sectional, its strong baseline is a cross-sectional mean rather than an own-routine R2, and its target is not tracked over time, so it fails the spine. Energy and markets are one track with two regimes, not two.

5 Evaluation Protocol and Pilot Plan

The scoring foundation is established practice we **adopt and cite**; we then add metrics for possibility-space resolution. ‘Collapse’ is the flagged epistemic metaphor of Section 1.3 — a distribution concentrating — with no physical meaning.

5.1 Preserved foundation (adopted)

Forecasts are scored with strictly proper rules: the **log score** (primary) and **Brier score** (secondary) [14,15,18]. The headline quantity is **Skill**, in bits:

$$Skill = \frac{\text{mean log-score}_{\text{system}} - \text{mean log-score}_{\text{reference}}}{\ln 2},$$

a paired comparison on identical sealed instances, equal in expectation to an information gain — not a novel metric. Skill is reported against **R1** (population prior, estimated leave-one-out without the scored target’s history and matched on question type, answer space, and horizon; the entry condition) and **R2** (the target’s recency-weighted, walk-forward routine; the target-specific condition). R2

is pre-registered, organizer-fit, and frozen (default recipe in Appendix C); it is intended as the strongest *simple* routine model and is admitted only where it itself beats R1, so ‘skill over R2’ cannot be manufactured against a weak baseline. **Calibration** gates the result [43], assessed by a pre-registered combination of calibration intercept/slope, a reliability decomposition, and top-label ECE bands (≤ 0.10 pass / ≤ 0.20 warn / > 0.20 fail). The **permutation specificity gate** scores a system’s model for instance i against another instance’s outcomes, matched within compatible question types, horizons, and answer spaces so that target identity is not confounded with base rates; skill that does not collapse was not target-specific. Of this stack, the R2 baseline and the permutation gate are the unoccupied elements; R1, proper scoring, calibration, and sealing are adopted from forecasting practice [14,15,18,28,40,43]. Scoring a distribution over a defined answer space also means exact surface form is not the success criterion — a principle the JEPA literature applies at the representation level (Section I.2) and we adopt at the level of scoring. Full reporting details — probability floors, stratified aggregation, recalibration policy, permutation-effect reporting and its power profile, and day-blocked / person-clustered uncertainty — are specified in Appendix J.

5.2 A worked instance (synthetic, illustrative)

To make the apparatus concrete — with invented numbers, no participant, and no result claimed — Figure 2 works through one instance. A single tracked person has a recurring Wednesday review on the calendar, but this week passive signals show attention redirected to an urgent dependency. The query is whether the review is completed by Wednesday 17:00 (answer space {completes, defers}). R1 (population prior), R2 (this person’s own routine), and the evaluated model assign $P(\text{defers})$ of 0.15, 0.10, and 0.70; the review is deferred, so the model gains about +2.2 bits over R1 and +2.8 over R2 — skill the routine did not contain. Scored against a *different* person who kept their review, the same forecast earns about −1.6 bits over that person’s R2, so its skill collapses and is confirmed specific to this target. A model that had learned only base rates or this person’s habit would match R1 or R2 and show no lift. (All numbers are illustrative.)

5.3 The control battery

TargetSpace is not only a dataset; it is an **evaluation protocol** for whether longitudinal target information improves prediction in a measurable way. A meaningful evaluation runs a model across a battery of conditions, each instantiating machinery defined earlier: a *zero-history* arm (operationally R1), a *short-history* arm (locating where added history first pays), and a *longitudinal-history* arm (the condition the benchmark rewards); a *shuffled-history* control (the correct target’s history in scrambled temporal order — a re-scored diagnostic that never alters a sealed forecast and isolates *order*) and a *wrong-target* control (the permutation gate, isolating *identity*); *ablated-modality* controls (the evidence-tier ablation); *oracle* and *human* anchors bounding the achievable range; *calibration*; and *evidence attribution* (the model identifies the observations supporting a forecast, enabling the explanatory audit a passed gate licenses). The three history arms plus the wrong-target gate suffice to substantiate the core claim; the remainder are recommended diagnostics. The shuffled-history and wrong-target controls separate the two ways a forecast can be target-specific in name only: indifference to *when* the evidence arrived, and indifference to *whose* evidence it was.

Ablation logic. The benchmark’s core claim is validated through these ablations, not asserted. *The benchmark’s central signal is not merely whether a model predicts the future, but whether prediction improves when the model is given the correct target’s history in the correct temporal order.* If target history, temporal order, target identity, or multimodal evidence do not improve performance, the benchmark is not measuring its intended capability, and the result should be

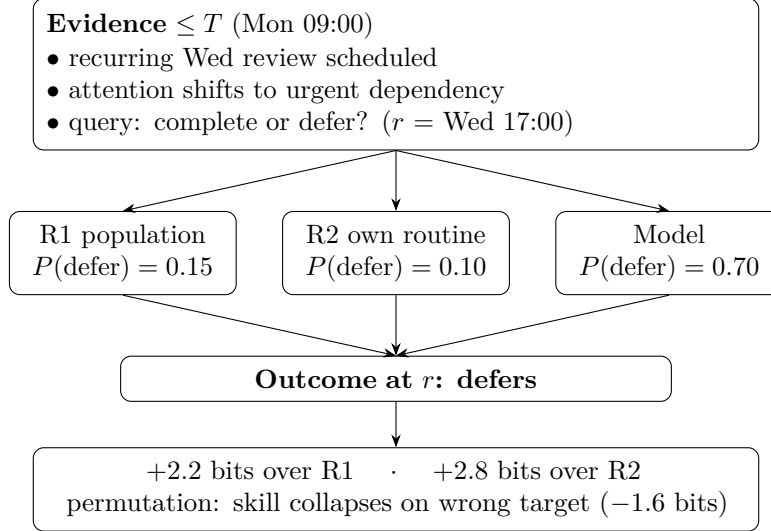


Figure 2: Worked synthetic instance of a target-state transition forecast. Evidence available at sealed time T supports three forecasts: a population baseline (R1), an own-routine baseline (R2), and the evaluated model. The outcome resolves later at r ; skill is scored against the sealed outcome, and a permutation check verifies that the forecast depends on the correct target–instance pairing. All numbers are illustrative and synthetic.

read as null. Conversely, if performance improves under longitudinal, target-specific conditions and degrades under the shuffled-history and wrong-target controls, the system is using target-specific dynamics rather than generic priors — exactly the inference the own-routine baseline (R2) and the permutation gate are designed to license (Section 5).

Standard baselines and ablation dimensions. The control battery pairs with a fixed set of baselines and ablations so results are comparable across systems. The **baselines** are: a *trivial-prior* baseline (R1); a *recency* baseline (recent-behaviour persistence, a component of R2); single-modality baselines — *calendar/schedule-only* (tier L0), *text-only* (L1), and *audio-only* (L2) — read off the evidence ladder (Section K.2); a *multimodal* model baseline; a *retrieval-augmented* baseline and a *long-context* baseline (both architecture classes, Section K.1); and a *human* or human-assisted baseline where feasible. The **ablation dimensions** are: *observation-window length* (the zero-, short-, and longitudinal-history arms); *modality removal* (the evidence-tier ablation); *target specificity* (the permutation gate); *memory / retrieval access* (toggling the store a system may query); *prediction horizon* (scoring the same sealed targets at multiple horizons, Section 2.2); and *temporal-leakage controls* (walk-forward sealing, Section 2.3). Each baseline and ablation is reported per evidence tier and per horizon, never pooled into a single headline number.

5.4 Reporting contract

TargetSpace is meant to be run, not only read. A researcher choosing to evaluate a system specifies, in order: **(1)** the target-system instances; **(2)** the forecast questions and their deterministic resolution rules; **(3)** the evidence tiers the system may access; **(4)** the architecture class(es) under test; **(5)** the R1 population-prior and R2 own-routine baselines; **(6)** the sealed forecast times and horizons; **(7)** the scoring metrics; and **(8)** the permutation test. The federated harness then runs where the data lives and reports a standardized row: **Skill vs R1**, **Skill vs R2**, calibration (pass/warn/fail), **permutation result** (collapses / survives), **predictive lift by evidence tier**, the **number of**

resolved forecasts, the **horizon**, and the **target domain**. The reporting contract is the point: every row discloses, simultaneously, whether the system beat the average, beat the routine, stayed calibrated, and depended on the correct target — so a reader can tell target-specific understanding from generic prediction at a glance.

5.5 Task tracks, pilot plan, and status

Version 1 uses auto-scorable, high-frequency targets organized into four domain-neutral task families instantiated for persons: **TS-R** short-horizon realization (next contact; event realization; response behaviour); **TS-A** attention/allocation; **TS-D** decision/commitment (commitment follow-through); **TS-X** target-state transition/drift. This paper reports no results. The first pilot is **harness validation**: a feasibility study of whether the protocol can run — forecast generation, sealing, deterministic resolution, variance and dependence estimation, evidence-tier feasibility, and preliminary signal over R1/R2. It does *not* test whether personal intelligence exists, does *not* validate cross-domain generality, and does *not* establish population-level effects. Concretely, it is an audio-first exploratory study: five consenting participants, thirty days, sealed daily calibrated predictions, systems differing only in evidence (A population prior; B digital exhaust; C +text/transcript; D +continuous first-person stream; model and prompts fixed across B–D). Its purpose is to check that the apparatus functions — forecast generation and sealing, reproducible outcome resolution, sufficient instance volume, and the variance and dependence structure of score differences — and to supply the inputs for *simulation-based* power planning of a later confirmatory study; with five participants it is not powered to confirm population-level effects, establish generalization, or estimate subgroup or cross-domain claims. A strong own-routine baseline compounds this: because a near-deterministic R2 leaves little headroom, detecting reliable lift over it may require high-frequency longitudinal observation and larger cohorts than the pilot provides. *Research hypotheses for the program* (tested confirmatorily only in a later powered study, and treated here as *exploratory measurements*): B–D beat R1 and R2 (H1); skill is non-decreasing in evidence, the open question being the C→D increment (H2); skill collapses under permutation (H3); null: no system beats R2 (H0). The primary endpoint is the paired prospective log-score improvement over R2, $\Delta L(M, R2) = L(R2) - L(M)$ in bits per forecast, with within-person day-blocked and between-person person-clustered uncertainty; performance over R1, calibration, and target-specific skill loss under matched permutation are validity gates. The attention-allocation family (TS-A) is the primary task family and commitment follow-through (TS-D) a key secondary; the final primary family and multiplicity rule are fixed in the pre-registered protocol. **Abandonment criteria**: no system beats R1 → noise; none beats R2 → target-specific forecasting not demonstrated; skill survives permutation → not target-specific.

Pre-registration and controls. Forecast instances are organizer-issued under a pre-registered eligibility and sampling frame — not entrant- or retrospectively selected — stratified across routine-continuation and transition cases and de-duplicated where evidence or outcome windows overlap; skill is reported separately on routine versus transition instances, since skill concentrated in transitions is stronger evidence of structure beyond routine replay. Outcomes are resolved by pre-registered rules: deterministic behavioural outcomes are preferred, ambiguous transition classes use forecast-blind adjudication with reported inter-rater reliability, unresolved cases follow a rule fixed before forecasts are examined, and an outcome label is never the same self-report channel a model uses as evidence. Missing evidence and unresolved outcomes may themselves be informative; their handling and sensitivity analyses are pre-registered, missingness is usable as evidence only when observable before the cutoff, entrants may not drop organizer-issued instances, and abstention is scored as a reference distribution under the same rule for human and machine systems. Negative controls comprise target permutation, a leakage / known-null canary, and a temporal-specificity (wrong-time-window) check.

Before any data collection the operational protocol pre-registers the R1 and R2 fitting, sampling frame, answer spaces, outcome-resolution rules, calibration split, permutation matching, missingness handling, abstention rule, primary endpoint, multiplicity rule, negative controls, and statistical analysis; full across-context, across-regime, and across-population transfer is later work, not a v1 gate.

6 Related Work and Positioning

6.1 What prior work already owns (and we adopt, not claim)

Credibility requires being explicit about what TargetSpace does *not* invent. It does not claim to invent prospective sealing, proper scoring rules, calibration evaluation, ablation analysis, or latent predictive models; its contribution is to assemble these controls around target-specific longitudinal forecasting, anchored on the own-routine baseline and the permutation-specificity gate. Appendix I (Table 10) itemizes each adopted component, its representative sources, and how TargetSpace uses it; Appendix I also carries the extended related-work discussion (memory and belief states, world models and predictive learning, forecasting and person modeling, cross-scale target-state systems, and intelligence measurement), of which this section keeps only the load-bearing comparisons. The nearest empirical neighbours — experience sampling [61], personal sensing [62], and digital phenotyping [64] — gather longitudinal individual-level data; person-modeling benchmarks [4,5,7,26] test retrospective comprehension or preference replication; goal recognition [29] identifies which fixed goal explains behaviour. TargetSpace differs in the evaluation object: a prospective, calibrated, instance-specific forecast of the *next* target-state transition, under R1/R2 and the permutation gate.

6.2 The conjunction

Each ingredient in Table 10 is occupied somewhere; the conjunction is not. We state the supported claim narrowly: **to our knowledge no existing benchmark scores prospective, calibrated, proper-scored forecasts of a tracked target system’s latent target-state transitions under a strong instance-specific own-routine baseline (R2) and an instance-permutation specificity gate, with an evidence-tier ablation, across architecture classes.** We do *not* claim to be first at architecture-neutral evaluation [41], cross-architecture comparison [42], calibrated sealed forecasting [28], evidence ablation, or resolution-timing [29]. The contribution is their assembly around one measurable question.

Table 5 audits neighboring benchmarks against the five dimensions. The labels are cautious design judgments, not numerical scores: *clear* (a primary, explicit target), *partial*, *arguable*, or *absent* (including dimensions that are simply not a primary evaluation target for that benchmark). Read across, prior work holds important *subsets* — sealed proper-scored forecasting (ForecastBench), a persistent individual (PersonaMem, Generative Agents, KnowMe-Bench), latent-goal inference (EgoToM, ToMnet, goal/plan recognition), novel-task adaptation (ARC-AGI). To our knowledge none combines all five in one prospective apparatus. The TargetSpace row reflects its proposed design, not completed empirical validation; the table is a structured literature comparison, not a measured result.

6.3 Positioning

Two caveats specific to this positioning (the general limitations are Section 7): TargetSpace is *not* a substitute for physical-reasoning, robotics, or video-realism benchmarks, and a strong score implies

Table 5: The five-dimension conjunction across neighboring benchmarks (1: persistent individual / evolving entity; 2: longitudinal, continuously updated evidence; 3: strictly prospective / sealed prediction; 4: calibrated proper scoring with meaningful baselines; 5: goal-state / meaningful-transition forecasting as the target). Labels are cautious judgments from each benchmark’s design, not measured scores. Prior work contains important subsets; we are not aware of a benchmark combining all five in one prospective evaluation apparatus.

Benchmark family	/ Persistent individual	Longitudinal evidence	Prospective / sealed	Calibrated baselines	+ Transition target
EgoToM [27]	absent	partial	partial	absent	arguable
ToMnet [6]	partial	partial	partial	absent	arguable
ForecastBench [28]	absent	absent	clear	clear	absent
PersonaMem [7]	clear	partial	absent	absent	arguable
Generative Agents [4,5]	clear	partial	absent	absent	absent
KnowMe-Bench [26]	clear	partial	absent	absent	absent
Goal / plan recognition [29]	partial	partial	arguable	absent	arguable
ARC-AGI [65,66]	absent	absent	clear	absent	absent
TargetSpace (this work)	clear	clear	clear	clear	clear

nothing about those capabilities; and models may overfit to retrieval-like cues — surfacing *what* happened — rather than learning target dynamics, which is why higher recall or longer context does not by itself move prospective skill over R1/R2 under the permutation gate (Section G.2). Predictability itself is bounded [13,91], so no single headline number can represent the construct; skill is reported per instance, against R1/R2, under the calibration and permutation gates.

Positioning statement. TargetSpace therefore occupies a complementary position in the world-model evaluation landscape. It does not attempt to replace benchmarks of intuitive physics, video realism, embodied control, or symbolic forecasting; it evaluates a missing layer — whether a model can transform passive longitudinal observation into a target-specific predictive model. This layer is central for deployed agents that must reason about particular people, teams, systems, projects, or environments over time. In this sense TargetSpace asks not only whether a model understands the world in general, but whether it can learn the dynamics of *this* target.

7 Limitations, Ethics, and Dataset Roadmap

What a positive result does and does not establish. A positive TargetSpace result establishes calibrated, prospective, target-specific predictive skill — not understanding in a rich sense, and not the superiority of any architecture. It begins the scientific question rather than ending it: once a system beats R1/R2 and fails permutation, *explanatory audit* of which evidence, constraints, attention shifts, or belief-state updates made the forecast possible becomes a disciplined second task (Section 2.2) — a post-hoc narrative earns credit only by improving later sealed forecasts or surviving pre-registered ablation. Conversely, skill measured under passive observation is predictive, not causal (an interventional design would be needed); the benchmark keeps attention-conditioned target formation, action-conditioned consequence prediction, and planning distinct (Sections 1.4 and 2.5); and a forecast can optimize a state while erasing the context that gives it meaning (engagement at the expense of well-being) — the permutation gate and the observe-not-intervene rule are partial guards, but detecting such context erasure in general remains open.

Dataset roadmap. The initial release includes no public raw first-person dataset; the synthetic harness (Appendix B) and the protocol are the released artifacts. A consent-governed passive-observation corpus — longitudinal first-person multimodal streams paired with sealed outcomes — is a core long-term objective, released only in stages under privacy, safety, de-identification, participant-control, and ethics-governance constraints, and possibly in centralized controlled-access, synthetic/de-identified, federated, or benchmark-server forms (Section 2.3).

Limitations, ethics, and deployment boundaries. (L1) Latent targets are evaluated only through observable proxies. (L2) The federated, prospective design makes reproduction harder than a downloadable dataset and precludes third-party audit of sealing, labelling, and resolution, so results are holder-reproducible, not publicly auditable; we mitigate with a versioned protocol and shared harness. (L3) Single-instance and small-cohort evaluation limits external validity. (L4) Predictability is bounded [13] and heterogeneous, so reporting is per-instance. (L5) Capture can be reactive — being observed may alter the behaviour being forecast — so habituation windows and reactivity checks are required, and induced deviations must not be read as target-state transitions. (L6) Domain-generalizability is a property of the formulation, demonstrated in one domain; the cross-domain claim is specified, not established. (L7) The privacy and governance safeguards are design commitments, not implemented features: no pilot has run, no institutional review has been obtained, and the privacy-filtering layer is unbuilt. (L8) A calibrated behaviour forecaster is dual-use, and the observe-not-intervene rule binds the benchmark, not downstream deployers. (L9) Longitudinal individual-level data carry selection and representation bias; a benchmark validated on a narrow cohort will not transfer, and per-instance reporting does not cure unrepresentative sampling. (L10) Ecological validity is limited: instrumented, consented capture may not reflect unobserved behaviour, and a habituation window does not guarantee it. (L11) Many target states admit annotation and label ambiguity — whether a commitment was ‘abandoned’ or ‘deferred’ can be genuinely contested — so resolution rules must be pre-registered and inter-rater reliability reported. (L12) Targets exhibit drift and non-stationarity: the dynamics a model fits can change, so R2 is re-fit walk-forward and skill decay is tracked rather than assumed stable. (L13) For some outcomes ground truth is intrinsically hard to establish — probabilistic, delayed, or only partially observed — which bounds achievable skill and is why scoring is proper rather than accuracy-based. (L14) Multimodal longitudinal capture is costly and burdensome to collect and store, constraining cohort size and biasing toward participants who can be instrumented. (L15) Results risk overfitting to a single person, household, or organization; the permutation gate guards against cross-target leakage but not against a model tuned to one idiosyncratic target, so external validity requires multiple independent targets. (L16) The benchmark measures forecasting of behaviour, not understanding of intent: a high score is predictive alignment with observable future states, and any inference about goals, reasons, or inner life is explanatory audit (Section 2.2) subordinate to the sealed forecast, never established by the score. We specify informed revocable consent, federation so raw data never leaves the target’s control, aggregate-only reporting, institutional review before any human-subjects deployment, respect for recording law, and a prohibition on evaluating systems that act on the target during the window — a benchmark that rewarded changing the target would measure influence, not understanding; bystander and third-party consent, non-anonymizable multimodal streams, withdrawal of already-sealed forecasts, manipulation, and surveillance normalization remain unresolved. We keep benchmark validity strictly separate from deployment legitimacy: a high score certifies only better prospective predictions about a consenting individual under sealed conditions — it grants no privileged access to a person’s inner life, is distinct from diagnosis, and confers no licence for employment, insurance, surveillance, coercive, or manipulative use. We model the lived trajectory only through its observable longitudinal traces and score only forecasts of future observable states; the framework claims nothing about subjective experience, qualia, or consciousness, and a high

score is evidence of predictive alignment with a target, not of access to a person’s mind. A system with rich observation and persistent memory must stay objective-bounded; a containment/refusal axis alongside the skill axis is required future work. A default minimal safe pilot configuration — consenting adults, local-first storage, export of sealed forecasts and aggregate metrics only, capture exclusions, and ethics review before recruitment — is given in Appendix D (Table 7, Figure 3).

7.1 The inference object, not the raw signal, is the privacy boundary

Privacy controls that focus only on raw audio or video retention are incomplete. Even if raw recordings are never stored or are deleted quickly, an ambient target-state system (Section 2) may retain transcripts, OCR, embeddings, scene labels, speaker and entity maps, commitments, routines, summaries, and inferred goals — derived objects that are often more searchable, more compressive, and more actionable than the recording they came from. *The privacy-sensitive artifact is not only the recording; it is the inferred experiential model* (Table 2). Deleting the raw recording does not necessarily delete the target-state risk if those derived layers persist. The same layering concentrates the other risks this section tracks: bystanders become data subjects (and visual capture extends this to faces, screens, documents, homes, and workplaces); purposes drift (meeting notes become behavioural prediction); compression preserves actionable facts while stripping context; forgotten remarks become queryable; and memory can be interrogated by whoever holds power over the target — employer, parent, platform, or legal adversary. Local processing and open-source implementations are valuable mitigations [96,101], not complete solutions to bystander consent, purpose drift, or derived-inference risk.

These systems also have clearly beneficial uses — accessibility, memory support, productivity, care coordination, personal knowledge management, safety — which is precisely why a shared measurement and governance vocabulary is needed rather than alarm: the benchmark makes *what longitudinal capture buys* a measured quantity, and the capture-to-inference ladder gives governance a boundary object other than the raw signal. The extended discussion, including the industry case cluster, is Appendix L.

Latent world models and transfer. Latent predictors are a natural architecture class here, and their open evaluation challenges are why TargetSpace scores calibrated distributions against sealed external outcomes rather than internal compatibility: latent state can be hard to ground, deterministic prediction may under-represent uncertainty, and long-horizon rollouts can drift [76]. A coherent internal rollout is not itself evidence of competence. Whether target-specific structure is reusable across regime shifts, beyond routine and memorization, is a transfer question the same bits-Skill and splits can measure (Section 7.2).

7.2 Open research questions

Each is a controlled comparison of target-specific Skill, grouped into five programs: **(1) evidence sufficiency** — which tier first beats R2, and which modalities add marginal skill; **(2) attention and target-state formation** — does attention-trajectory evidence add calibrated lift beyond routine and stated goals, and how early can an emerging target be detected before explicit commitment; **(3) transfer** — does target-specific structure stay predictive across routine-breaking and withheld windows, distinct from memorization, and how fast does skill decay once evidence stops; **(4) architecture and horizon** — do explicit consequence models calibrate better than reactive policies, and at what horizon do detailed rollouts lose to abstract target-state forecasts; **(5) reactivity and active evidence acquisition** — can target-state formation be separated from observer-induced reactivity, and can a system request more evidence without overreaching, as a measured, consent-

aware capability. Recent work frames reusable, computationally-bounded structure as *epiplexity* [80]; we note it as related framing but do not adopt it as a metric, since estimating it needs a reference learner’s training dynamics that closed APIs and human baselines do not expose, whereas bits-Skill already reports the prospective predictive code-length gain architecture-neutrally.

8 Conclusion

Many systems now claim to understand specific targets, and almost none can show their forecasts are target-specific rather than reflections of base rates and routine. TargetSpace makes that distinction measurable: prospective, sealed, proper-scored forecasting of target-state transitions under partial observation, certified by an own-routine baseline, a calibration gate, and a permutation specificity test, with an evidence-tier ablation and an architecture-neutral grid. Its novelty is the conjunction assembled around one question — *is the forecast about the target, or about the average?* — not any single ingredient, each of which it adopts and cites. Personal world modeling is the flagship instantiation, and TargetSpace scores the external forecast, not the architectural story, whether a system is reactive, predictive, or hybrid. It begins from one premise: the target was never a profile, a preference list, or a memory store, but a lived trajectory — an unfolding stream of attention, pressure, constraint, and emerging goals — which the benchmark asks a model to infer from observable traces and to forecast where it turns next.

Forecast the target, not the average — and model how the target is being formed, not only which label best describes it.

The target is not a profile. The target is a lived trajectory.

Declarations

Author contributions. Yuri Andrade Sylvester conceived the TargetSpace framework, developed the benchmark specification and proposed evaluation protocol, conducted the literature synthesis, and wrote and revised the manuscript.

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Ethics statement. No human-participant data were collected for this paper. Dataset collection has not begun, and no ethics approval has been sought or obtained for the proposed future study. Any future participant research implementing this protocol will require appropriate ethics review and informed consent before recruitment or data collection.

Data availability. No participant dataset was generated or analyzed for this paper.

Code availability. A minimal synthetic demonstration harness accompanies this preprint as an ancillary file (Appendix B); it uses no participant data and supports no empirical claims. The full reference implementation and participant-data harness remain in preparation.

Project website. The project website is in preparation and is not cited in this version.

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A Glossary of core terms

Definitions are deliberately compact; the body is authoritative where they differ.

Target state. A latent configuration a target behaves as if acting to reach, maintain, resume, or abandon; evaluated only through observable consequences, never scored directly. **Stated goal.** What a target declares it wants — evidence, not a scored quantity. **Enacted priority.** What a target’s allocation of a scarce resource (attention, time) reveals, independent of what it states. **Inferred goal.** What a model concludes a target is pursuing; evidence-bearing latent structure, scored only via a mapped outcome. **Implicit target.** A latent orientation inferred from behaviour, repetition, avoidance, or attention that the target has not declared and may not introspect. **Score target.** A pre-registered discrete future state or transition with an externally observable resolution — the only object that earns or loses credit. **Sealed outcome (outcome).** The realized value of a score target at the resolution time, fixed by a deterministic rule and hashed before it exists; scoring compares the sealed forecast against it. **Forecast horizon.** The interval $h = r - t$ from the sealed forecast time t to the resolution time r ; forecasts are specified and scored per horizon. **Transition window.** The span over which a target-state transition is deemed to occur for resolution purposes, distinct from the forecast horizon. **Belief state.** The observer/model’s distribution over a target’s latent variables, updated as evidence arrives; a sufficient statistic of the history that the benchmark

motivates but never scores. **Transition.** A change in which target state a system is acting to reach: emergence, stabilization, competition, displacement, abandonment, resumption, or opportunity-induced creation. **Population-prior baseline (R1).** A reference that predicts from population base rates with no target-specific information; the entry condition a model must beat. **Own-routine baseline (R2).** A pre-registered, walk-forward model of the target’s own recent routine; skill over R2 is the headline target-specific signal. **Evidence tier.** A cumulative band of the evidence ladder (Section K.2) describing which streams a system may use, from low-content metadata to specialized sensors. **Evidence ablation.** Measuring skill over R2 as a function of evidence tier, to test which streams add target-specific information. **Permutation specificity gate.** A test that scores a system’s forecasts for one target against another target’s outcomes; genuine target-specific skill should collapse. **Prospective sealing.** Timestamping and hashing (SHA-256) a forecast before its outcome exists, preventing hindsight and leakage. **Walk-forward evaluation.** Prequential scoring using only evidence timestamped at or before t ; random cross-validation is prohibited. **Calibration gate.** A pass/warn/fail check that stated confidence tracks empirical frequency (proper scoring plus expected-calibration-error bands). **Target-specific skill.** Calibrated predictive skill, in bits, that exceeds both R1 and R2 and collapses under permutation. **Epistemic resolution.** The falling of the observer’s uncertainty as evidence accumulates (posterior concentration), distinct from participatory formation in the target. **Participatory dynamics.** Observable feedback, attention, action, and constraint signals through which a target’s own process may change which futures remain reachable; recorded as auxiliary diagnostic variables (Appendix F), never part of primary scoring.

B Minimal synthetic harness

Purpose. The harness is a sanity check of the evaluation machinery, not empirical evidence that passive observation improves personal forecasting. It exercises the input and output schemas, the two baselines, the scoring rules, and the diagnostic gates end to end on data we generate ourselves, so that a reader can confirm the pipeline runs and produces the reported quantities. The shipped artifact is a single deterministic, pure-Python-stdlib script distributed as an arXiv ancillary file (`anc/targetspace_synthetic_demo.py`). The multi-file layout below is the planned structure of the fuller reference implementation, in preparation.

What it demonstrates. Synthetic target histories; walk-forward forecast instances; the R1 population-prior baseline; the R2 own-routine baseline; a simulated model forecast; log (in bits) and Brier scoring; a calibration diagnostic; a permutation specificity check; and evidence-tier reporting.

What it does not demonstrate. No human data; no real personal intelligence; no evidence that passive observation helps; no cross-domain validation; no safety validation.

Planned file layout (reference implementation, in preparation).

- `README.md`
- `generate_synthetic_targets.py`
- `run_walk_forward_eval.py`
- `baselines.py`

- `scoring.py`
- `permutation_test.py`
- `example_config.yaml`
- `synthetic_results_example.csv`
- `leaderboard_example.csv`

Forecast-instance (input) schema. `instance_id`, `target_id`, `forecast_time`, `resolution_time`, `question_type`, `answer_space`, `evidence_tier`, `available_features`, `resolved_outcome`, `split_or_fold`, `admissible_for_R2`.

Forecast-output schema. `forecast_id`, `system_id`, `target_id`, `forecast_time`, `question_type`, `answer_space`, `probability_vector`, `evidence_tier`, `model_access_mode`, `sealed_hash`, `timestamp`, `adapter_version`.

Example commands. `python generate_synthetic_targets.py -config example_config.yaml ; python run_walk_forward_eval.py -config example_config.yaml ; python permutation_test.py -predictions outputs/predictions.csv.`

Example leaderboard row. Values are illustrative synthetic outputs of the demo script; they are not empirical results.

Table 6: Illustrative synthetic leaderboard row produced by the demo script. Skill is measured in bits over the named reference baseline.

<code>system_id</code>	<code>tier</code>	<code>vs_R1</code>	<code>vs_R2</code>	<code>brier</code>	<code>cal. warn</code>	<code>perm. loss</code>	<code>n_tgt</code>	<code>n_fc</code>
toy-L2	L2	+0.049	+0.036	0.183	none	collapses	20	800

The columns report, in order, the system identifier, evidence tier, skill in bits over R1, skill in bits over R2, Brier score, calibration-gate warning status, the permutation specificity outcome (skill collapses under permutation, as required), and the target and forecast counts.

C Default R2 own-routine baseline specification

This appendix fixes a pre-registered default recipe (v1) for the R2 own-routine baseline. The recipe is provisional and protocol-configurable: the values below are defaults to be frozen before evaluation, not claims about any observed data. The intent is to make R2 a fair, routine-only competitor against which target-specific skill in bits can be measured without manufacturing lift against a weak baseline.

Admission. R2 is admitted for a given target and question type only when two conditions hold jointly: the target has enough prior resolved history (see minimum history below), and R2 beats R1 on a pre-seal validation slice (or a rolling historical criterion fixed before the seal). When R2 does not beat R1 on that slice, we report that R2 is not informative for that stratum and fall back to R1 as the reference; we do not manufacture target-specific lift by scoring against a deliberately weak own-routine baseline.

Minimum history. The default minimum is the stricter of 20 prior resolved opportunities or 14 days of eligible history per target and question type. Strata that do not meet this threshold are not admitted for R2 and are reported separately. The threshold is provisional and protocol-configurable.

Features (allowed). R2 may use only simple pre-forecast routine features: marginal historical frequency for the target $\times$ question type; recency-weighted frequency; persistence or previous target state; recent target-state transition counts; time-of-day; day-of-week; and known recurring calendar or deadline commitments that are available before forecast time.

Exclusions. R2 may not use any of the following: passive audio, video, or screen content; semantic text content beyond pre-specified routine labels; any future information relative to the evidence available up to time t ; entrant model outputs; manually selected post-hoc features; or any representation learned from the scored test outcomes.

Fitting. R2 is organizer-fit and frozen before evaluation. Fitting is walk-forward only: no random cross-validation and no tuning against test outcomes. Parameters are estimated from history available before each forecast time and never revised using resolved test labels.

Smoothing. R2 applies a pre-registered smoothing scheme and a nonzero probability floor over the answer space A , so that every admitted answer receives positive probability and proper scores (log score in bits, Brier) remain finite. Exact smoothing constants and the floor are protocol parameters fixed before test results are seen.

Reporting. Skill in bits is reported against R1 and against R2 separately, never collapsed into a single number. Strata in which R2 fails admission are reported as such, with R1 as the reference. Incomparable answer spaces are not pooled without strata.

D Safe pilot configuration

The benchmark can be tested without assembling an uncontrolled lifelogging dataset. Table 7 lists a default minimal safe pilot configuration, and Figure 3 shows the federated, local-first architecture it assumes: raw capture stays on the participant device, and only sealed forecasts, resolved labels, and aggregate metrics leave the client.

E Supplementary figures

These figures restate, in visual form, material presented as tables in the main text: Figure 4 corresponds to the target-specificity stack of Section 3, and Figure 5 to the evidence ladder of Section K.2.

F Participatory dynamics as auxiliary state variables

TargetSpace keeps epistemic scoring separate from target-state formation. Forecast accuracy is scored only against sealed external outcomes (Section 2). Separately, the harness can record observable feedback, attention, action, and constraint signals as *auxiliary* variables, letting researchers test whether adaptive feedback loops change the reachable target-state space and whether those changes

Table 7: Minimal safe pilot configuration. The benchmark can be tested without creating an uncontrolled lifelogging dataset. The following practical controls are the default for any pilot.

Default controls for a minimal safe pilot

- Consenting adults only.
 - Local-first storage.
 - No public release of raw video or audio.
 - No raw third-party export by default.
 - Only sealed forecasts and aggregate metrics leave the device or client.
 - Participant controls for review, pause, deletion, and opt-out.
 - No capture in bathrooms, bedrooms, medical or legal settings, schools, children’s spaces, or confidential meetings.
 - Visible recording notice where feasible or required.
 - Bystander redaction or exclusion where feasible.
 - Short raw-media retention window unless explicitly consented otherwise.
 - Ethics / IRB (or equivalent) review before recruitment.
 - Special handling for third-party content, employee contexts, children, and sensitive data.
-

improve forecasts. Two processes must not be conflated: *epistemic resolution* is the observer’s uncertainty falling as evidence accumulates — posterior concentration over the observer’s maintained distribution; *participatory target-state formation* is a change in the target itself — through attention, action, feedback, and constraint — that alters which futures remain salient, reachable, reinforced, or stable. Posterior concentration in the observer is not target-state formation in the observed system; the benchmark continues to score forecasts only against sealed outcomes, and the proxies below are a diagnostic layer, not part of the scoring foundation.

- *Feedback exposure index*: count, source, valence, intensity, latency, and channel of feedback the target receives.
- *Attention shift index*: change in time, dwell, mentions, or returns to objects of attention before versus after feedback.
- *Action-coupling index*: whether attention changes are followed by observable actions — emails, calendar changes, task completion, purchases, commitments, code commits, document edits.
- *Constraint / reachability map*: whether resources, obligations, blockers, commitments, permissions, or pathways are added or removed.
- *Forecast sensitivity audit*: compares passive-only, feedback, feedback+attention, and feedback+attention+action/constraint feature sets, testing whether participatory variables improve sealed-outcome forecasts.

A positive finding — forecasts improving when participatory variables are added, and the reachable target-state space measurably changing under feedback — would be prospective evidence that adaptive loops matter, not an assumption. A null finding leaves the primary scoring unaffected.

The attention–target dynamical loop (extended treatment)

The main text (Section 1.4) treats attention as evidence and as a candidate source of predictive lift. The underlying cognitive-dynamical reading is a loop rather than a one-way law: repeated allocation can reinforce and stabilize a provisional objective; withdrawal can weaken, delay, or displace one;

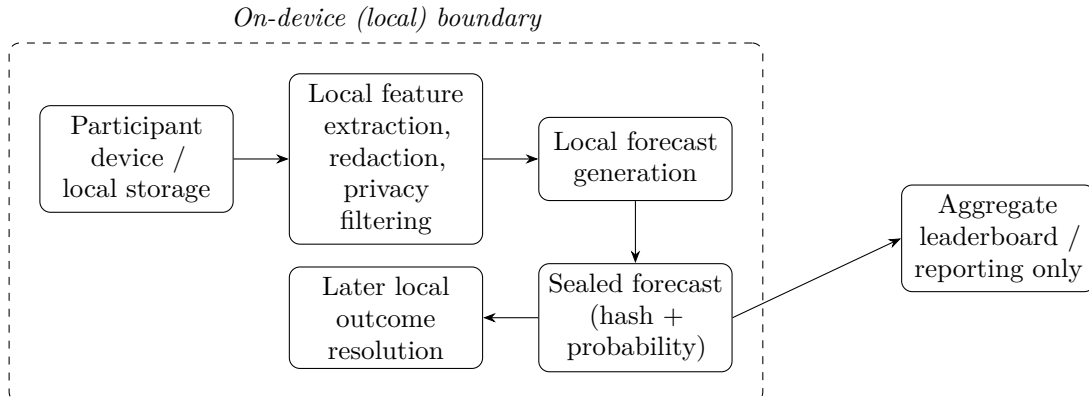


Figure 3: Safe federated pilot architecture. Raw audio, video, and screen data stay on the participant device: feature extraction, redaction, privacy filtering, forecast generation, sealing (hash and probability), and outcome resolution all run locally within the dashed on-device boundary. Only the sealed forecast (and, later, resolved labels and aggregate metrics) leaves the client; the off-device stage performs aggregate leaderboard and reporting computation only. No raw capture is transmitted.

newly encountered evidence or opportunity can make determinate a target that previously was not [72,73]. Schematically,

$$z_t \rightarrow a_t \rightarrow (e_t, u_t) \rightarrow z_{t+1},$$

where z_t is the latent target state, a_t attention allocation (or an observable proxy), e_t newly encountered evidence, and u_t action. Attention is thereby both evidence of the current target state and one candidate process through which the next is formed. Prior dispositions, habit, environment, and constraint remain primary; attention does not create targets from nothing; and the loop is exploited by the benchmark only as a source of falsifiable predictive-lift hypotheses, never as an assumed mechanism.

Extended motivation: experience, not knowledge

The bounded ambition behind the personal track is to model not what a person *knows* but — in a precise and deliberately limited sense — what it is like to be that person, borrowing the phrase only as a motivating contrast between third-person description and situated experience [92]: not inner experience, to which a system has no access and about which no claim is made, but the lived trajectory from which behaviour emerges — the unfolding, situated stream of attention, context, pressure, memory, constraint, and possibility that shapes what becomes salient and what the person does next. Knowledge is episodic and declarative; experience is longitudinal, situated, relational, embodied, multimodal, and predictive. TargetSpace evaluates the transition from passive experience capture to target-state inference, and scores it only through externally resolvable forecasts.

G Personal track: task specification and extended discussion

To show that TargetSpace instantiates runnable tasks and not only a framework, we specify one minimal personal-track task, the *Recurring-Commitment Completion Forecast*. It is synthetic and illustrative; no empirical results are reported.

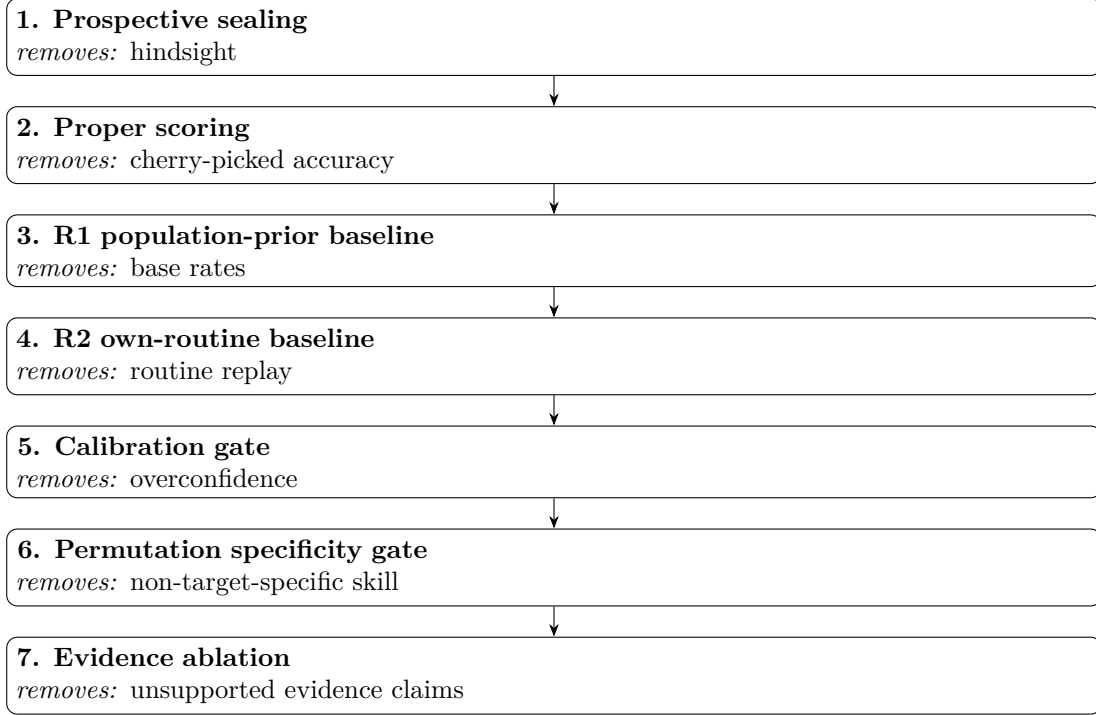


Figure 4: The target-specificity stack, complementing the stack table (Section 3). Each layer removes one way a forecast can appear target-specific without being so: a forecast that survives all seven filters cannot be explained by hindsight, selective reporting, population base rates, replay of the subject’s own routine, overconfidence, non-target-specific skill, or unsupported evidence claims. Layers 4 and 6 (the R2 own-routine baseline and the permutation specificity gate) are the anchors contributed here.

- *Task.* Given timestamped evidence available up to a sealed time T , forecast how a scheduled recurring commitment resolves by a later resolution time r .
- *Inputs (by tier).* Calendar and task history, communication metadata, and prior completion/defer/cancel history (L0); optional user-authored text traces (L1); optional passive-observation summaries (L4). See Section K.2.
- *Answer space.* $A = \{\text{complete, defer, cancel, replace}\}$.
- *Baselines.* R1 (population prior over the four outcomes) and R2 (the target’s own recurring-commitment routine), both fit walk-forward on evidence before T .
- *Resolution.* A deterministic rule over later observable evidence — calendar status, attendance, task completion, or a pre-registered behavioural marker — fixes the outcome at r .
- *Scoring.* Log score (bits) and Brier against the sealed outcome; report skill over R1 and over R2 (Section 5).
- *Specificity.* A permutation test across matched targets should reduce or collapse the skill if a model’s forecast was genuinely target-specific.

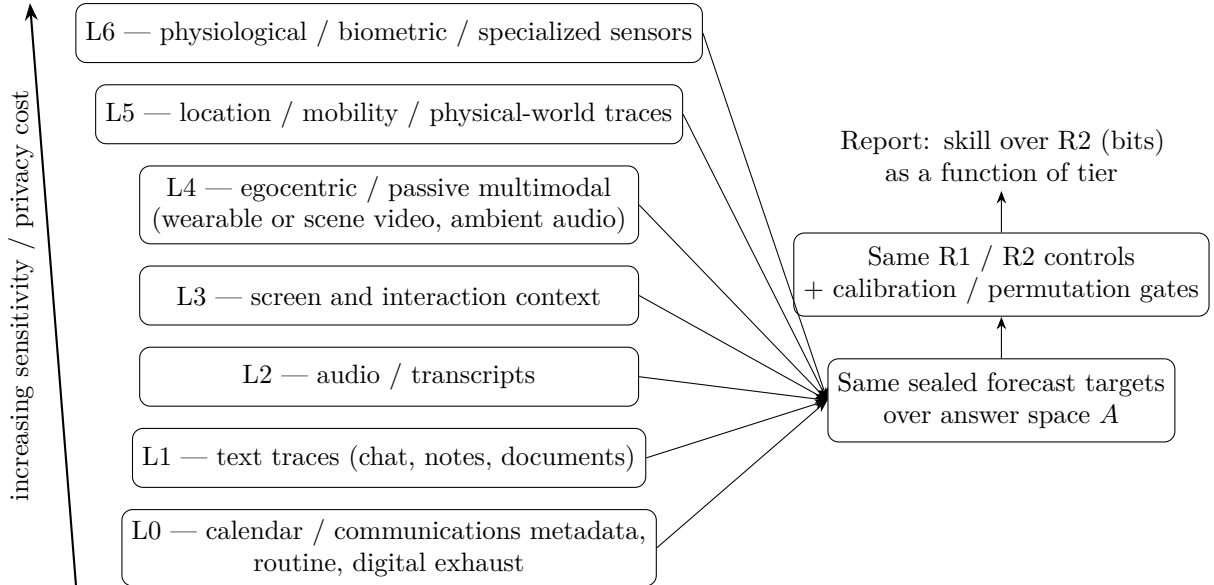


Figure 5: Evidence-tier ablation design (Section K.2). Seven evidence tiers (L0, least sensitive, at bottom; L6, most sensitive, at top) are each evaluated against the *same* sealed forecast targets and the *same* R1 (population-prior) and R2 (own-routine) controls under the calibration and permutation-specificity gates. Whether richer evidence helps is measured as skill over R2 (in bits) as a function of tier, not assumed; sensitivity and privacy cost increase upward.

The task is small enough to run on current language models, agents, and retrieval or memory systems, and exercises the full stack: sealed prospective scoring, R1/R2 baselines, calibration, evidence tiers, and the permutation gate.

Horizon–abstraction hierarchy

The four task families are scored across the following horizon–abstraction hierarchy.

Table 8: A horizon–abstraction hierarchy for target-state forecasting. Each level is scored independently; no single representation is assumed optimal across horizons, and resolution rules are deterministic or pre-registered. Uncertainty generally grows with horizon.

Level	Horizon	Answer space / resolution rule	Baseline; metric
Immediate action / next state	seconds–minutes	concrete next state; observed action	R2; log/Brier, calibration
Short-horizon task transition	minutes–hours	small discrete set; observed completion or switch	R2; Skill, top- k
Medium-horizon sub-goal / commitment	hours–days	commitment set; follow-through within a window	R2; Skill, time-to-resolution
Longer-horizon priority / target-state shift	days–weeks	priority/regime set; sustained reallocation of attention/resources	R1/R2; transition-path likelihood
High-level regime / strategy transition	weeks+	coarse regime labels; observed regime change	R1; Skill, calibration

G.1 The corpus bottleneck and the capture-to-labels stack

The missing data layer named in Section 1.3 is concrete in this domain. Digital exhaust (L0–L1) records a person’s *outputs* — what they wrote, sent, or scheduled — not the process that produced them. Passive first-person observation (L2–L4) is the stream from which goals, constraints, interruptions, and transitions could be inferred *before* intent is formalized: passive audio, a screen / digital-activity timeline, documents and files touched, calendar and task context, interruptions, spoken intent, environmental and social context, and the timing and sequencing of actions. Egocentric corpora such as Ego4D and Ego-Exo4D [56,57] show passive first-person capture at scale, but as generic clips for activity recognition — not longitudinal records of one tracked individual aligned to that person’s goals and outcomes over time.

Raw capture is not, by itself, a benchmark. Turning observation into evaluable state-transition data takes a stack — passive sensing, a longitudinal multimodal corpus, segmentation with goal/task/outcome/transition labels, a temporal state representation, the prediction benchmark, and a privacy-filtering layer (Section 7). **TargetSpace is the evaluation layer:** it fixes the evaluable target and the scoring spine and specifies what the upstream layers must deliver, but does not supply the corpus. Assembling a labelled, privacy-filtered, goal-aligned longitudinal corpus is the open bottleneck, and the pilot (Section 5.5) builds only a thin slice; passive recording alone is insufficient. For high-sensitivity tiers (L2–L6), federation is necessary but insufficient: a viable implementation would likely require on-device filtering that converts raw audio, video, and screen streams into task-relevant semantic events before persistence — mitigating, though not solving, the bystander-consent and recording-law problems of Section 7. **Hardware is the capture layer, not the benchmark:** wearable and passive sensors raise stream fidelity, but TargetSpace takes no position on a device; whether richer capture pays is the evidence-tier ablation (Section K.2). A human-centered world model must be learned from observation, since language and digital exhaust alone are too sparse and post-hoc.

G.2 TargetSpace as a benchmark for human-centered world models

The world-model research direction defines intelligence operationally as the ability to predict the consequences of actions in an abstract representation space and to plan over those predictions [33,36]. TargetSpace applies that idea to human-assistive AI. The ‘environment’ is not a physical system but the latent state of a person: can a system infer, from passive observation up to a sealed time T and *before the user writes an explicit prompt*, the user’s current goal state, hidden constraints, the likely next transition, and the next useful action? It is a world model whose dynamics are a human task process, scored by calibrated skill over R1/R2 rather than by reward or reconstruction.

A behavioral world model, not a physical one. The world-model idea (above) earns its power because predicting the consequences of actions lets an agent test a plan before acting [12], with planning as search over predicted futures [37,38]. TargetSpace generalizes this predictive substrate, not its physical content: it asks for no prediction of torques, contacts, or pixel dynamics and claims no understanding of a physical world it never observes. The classical form is *state + action* $\rightarrow$ *next state*; the TargetSpace form is *passive longitudinal observation* $\rightarrow$ *belief state over latent targets* (Section 2.2) $\rightarrow$ *target-relevant future behaviour*. A competent system must therefore learn a compact behavioral world model of one observed person — equivalently a personal, or target-state belief, model — capturing recurring contexts, constraints, stable preferences, salient episodes, unresolved decisions, and target transitions, and predicting how that target state evolves as evidence arrives. The analogy is to the *form* of the predictive state, not its physical realization; like any learned dynamics model it can compound error over long horizons [76], and only its sealed

forecast is scored — not planning, action, or the model itself (Sections 2.5 and 7).

The shift is clearest by contrast with what existing evaluations ask. A **transcript** or chat benchmark asks *what was said?*; a **memory** benchmark asks *what happened?*; TargetSpace asks a third question — *what goal state is the person in, what transition is likely next, and what intervention would help or harm?*

Retrieval and long context are not the capability. Two now-standard memory paradigms make the same point concrete. **Retrieval-augmented generation** conditions a model on documents fetched from an external store [90]; **long-context** models simply hold more of the history in the window. Both surface *what happened* more readily — usefully — but neither is scored on whether the system inferred *why* an episode mattered or *what target* it implies: a retrieval system can surface the reread email, whereas TargetSpace asks whether a model reads it as an avoided decision and forecasts the slip. That is why a higher recall@ k or a longer context window does not by itself move prospective skill over R1/R2 under the permutation gate; the architecture grid (Section K.1) accordingly places base, long-context, retrieval, episodic-memory, and belief-state-updating systems on identical sealed instances.

Prompts may reveal goals but often omit the **pre-prompt** process — a prompt is a late, compressed artifact emitted *after* an intention has partly formed — which passive observation can preserve (attention, interruptions, partial actions), and the evidence-tier ablation tests whether preserving it improves prediction. TargetSpace does not build this user-centered world model; it operationalizes it as evaluation: fixing the latent quantities to predict, the baselines to beat (R1/R2), and the calibration and permutation gates that separate a genuine model from a fluent guess. The target is abstract predictive state, not raw reconstruction, and the framing is architecture-neutral (Section K.1, Table 11).

G.3 Asymmetric observability: self-report, observation, and person knowledge

Personal intelligence forces a question the benchmark must handle carefully: who models the person better — the person, or an outside observer? We reject any claim of observer superiority. An individual has privileged access to immediate subjective experience, private beliefs, felt emotions, conscious intentions, and autobiographical meaning. A longitudinal observer can sometimes detect repeated behavioural regularities, inconsistencies between stated and enacted priorities, avoidance patterns, cross-situational structure, and slow changes that are hard to notice from inside the stream of experience. Self-report and observation are therefore partially overlapping but non-redundant evidence channels. **The issue is not observer superiority, but asymmetric observability.** Their relative predictive value is an empirical question the benchmark should measure, not assume.

This mirrors person-perception research. The self-other knowledge asymmetry model holds that self and informants know different things — the self more accurate for internal, low-observability traits, informants for evaluative, highly observable ones [58] — and meta-analyses find informant reports valid and often incremental to self-report, for some outcomes exceeding it [59,60]. Self-report has limits of its own: people often lack introspective access to what drives their behaviour [23], which motivates real-time experience sampling [61] and passive ‘personal sensing’ [62,64]; internal and external self-awareness are themselves distinct constructs [63]. None of this makes observation more accurate than self-report — it makes the two complementary.

TargetSpace makes the comparison *prospective and empirical* rather than rhetorical. Self-report is an evidence modality and an architecture class (human self-report), and a baseline — not a gold standard; passive observation is another modality, not automatically more accurate. The evidence-tier ablation, and the head-to-head of human self-report against observational systems on identical sealed instances, measure which modality — or which combination — yields more

calibrated, person-specific skill for a given target. We expect a structure rather than a winner: **first-person access may be richer locally, while longitudinal external observation may be more informative structurally for some prediction targets**, and combinations of modalities may outperform any single one. The benchmark is built to find out, not to presume; clinical observation is the right analogy — complementary evidence, not clinician omniscience.

G.4 Implicit Targets and Self-Narrative

Surface action is an ambiguous signal of what a person is organizing around, so it helps to separate three layers. The **action layer** is what the person does (rereads a message; reopens a document; revisits a calendar event). The **narrative layer** is how they frame it (“I’m just making sure I understand it”). The **target layer** is what the behaviour is actually organizing around — which may be avoiding a difficult reply, reducing uncertainty before a decision, preserving a relationship, or protecting status. TargetSpace is primarily a benchmark for the third layer, using the first two as evidence. The separation is not decorative: declared intentions predict behaviour only weakly (the intention–behaviour gap [84]), motives inferred from behaviour diverge systematically from self-attributed ones [85], people may not introspect the causes of their own behaviour (Section G.3), and self-narration — inner speech that mediates attention, monitoring, and self-regulation [89] — can report a goal other than the one the longitudinal pattern reveals. A model that predicts only the action layer, or that trusts the narrative layer at face value, will miss the target layer exactly where it matters.

This is where surface behaviour becomes informative only longitudinally. Considered once, rereading an email, drafting but not sending a reply, repeatedly checking a thread without responding, or shifting attention as a deadline nears are each individually ambiguous. Across a tracked history they can mark a latent target under tension — an unresolved decision, an avoidance pattern, a commitment about to slip — precisely the deviations from routine where skill over R2 is earned. The benchmark does not ask a model to *narrate* this latent state; it asks it to convert these traces into a calibrated forecast of a pre-registered observable outcome (Section 2.2) and certifies, through R2 and the permutation gate, that the resulting skill is specific to this person rather than to population-level patterns of attention or delay. The cognitive-science framing here is motivation, not mechanism: TargetSpace takes no position on theories of inner speech or self-narrative and stands if they are revised — it requires only that some behaviourally relevant targets are implicit and inferable from passive longitudinal observation, which the evidence-tier ablation then tests rather than assumes.

H Transition taxonomy and resolution rules

The full taxonomy of target-state transitions the personal track forecasts (referenced from Sections 2.1 and 4.2) is given in Table 9. Each transition type carries a deterministic or pre-registered resolution rule grounded in later observable consequences, an ambiguity/evidence profile, and the baseline expected to be hardest to beat.

I Adopted components and extended related work

Table 10 itemizes the components TargetSpace adopts from prior work, their representative sources, and the use each is put to; the two anchors contributed here are the R2 own-routine baseline and the permutation specificity gate (Section 3).

Table 9: A taxonomy of target-state transitions the personal track forecasts. Each has a deterministic or pre-registered resolution rule grounded in later observable consequences, not subjective report. “Likely baseline” is the baseline expected to be hardest to beat; “horizon” is the typical forecast range.

Transition type	Operational description / resolution criterion	Ambiguity risk; evidence needed	Baseline; horizon
Routine continuation	target persists; resolved by continued allocation/action matching the prior pattern	low; digital exhaust suffices	R2; short
Latent-but-active target	already pursued but unstated; resolved by later action or commitment	medium; behaviour + context	R2; short-medium
Emerging target	a new target becomes determinate; resolved by first commitment or resource expenditure after onset	high (onset timing); attention + context	R1/R2 medium weak;
Stabilization	a provisional target consolidates; resolved by repeated allocation across windows	medium; attention trajectory	attn-conditioned; medium
Competition	two targets contend; resolved by which receives sustained allocation	high; attention + outcomes	attn-conditioned; short-medium
Displacement / priority reversal	a new target supplants the prior one; resolved by a switch in sustained allocation	medium; attention + calendar/task	attn-conditioned; medium
Abandonment	a target is dropped; resolved by absence of follow-through past a window	medium; non-action over time	R2; medium
Resumption	a prior target returns; resolved by renewed allocation after a gap	high; longitudinal history	R2; medium-long
Constraint-forced transition	a constraint forces a change; resolved by the observed transition under the constraint	low-medium; constraint + outcome	R1; short
Opportunity-induced target	new evidence or opportunity creates a target; resolved by uptake after the encounter	high; evidence event + uptake	R1/R2 short-medium weak;

I.1 Memory, belief states, and goal inference

Three literatures bear on what TargetSpace asks a system to maintain, and clarify what it does *not* reward. **Memory.** Cognitive science distinguishes episodic from semantic memory [83] and, more pointedly here, treats autobiographical memory as a goal-driven *construction* rather than a stored replay [81], with the same constructive system used to simulate the future as to reconstruct the past [82]. This is why we frame the target capability as forward-looking belief maintenance rather than archival recall (Section 1.3). **Belief states.** Maintaining a distribution over latent variables, updated from partial observations and serving as a sufficient statistic of the history, is the belief-state formalism of partially observable decision processes [88]; TargetSpace adopts the formalism as motivation (Section 2.2) but scores only the externally resolved forecasts a belief state produces. **Latent-goal inference.** That observers recover latent goals and beliefs by inverting a model of approximately rational action is the Bayesian theory-of-mind / inverse-planning account [86,87], the cognitive-science complement to the machine theory-of-mind [6] and goal-/plan-recognition [29] work discussed above; behaviour is further organized by implicit motives that diverge from declared ones [85] and by intentions that predict action only weakly [84]. TargetSpace differs from all three in object and protocol: it scores neither recall (memory benchmarks), nor the internal belief state (world-model evaluation), nor which goal from a fixed set best explains past behaviour (goal recognition), but the prospective, calibrated, instance-specific forecast of the next target-state

Table 10: Prior components and TargetSpace-specific contributions. TargetSpace adopts prospective sealing, proper scoring, calibration, evidence ablation, and latent-predictive evaluation from prior work; its own contribution is target-specific forecasting under strong controls — the R2 own-routine baseline and the permutation specificity gate — assembled around one evaluation question.

Prior component	Representative sources	TargetSpace use
Architecture-neutral world-model evaluation	AutumnBench / WorldTest [41]	Adopt; add calibration, R1/R2, permutation
Cross-architecture comparison	WorldPrediction [42]	Adopt; add proper scoring, sealing, specificity
Latent predictive learning	CPC [44]; JEPA [33–36]	Evaluate as an architecture class
Proper scoring & calibration	Good; Brier; Gneiting & Raftery [14,15,18]; HELM [43]	Adopt directly
Sealed prospective forecasting	ForecastBench [28]; Prophet Arena [40]; SWE-bench [2]	Adopt
Evidence ablation	digital phenotyping / personal sensing [62,64]	Adopt; report as lift over R2
Goal recognition & resolution timing	goal-recognition design / WCD [29]	Adapt to sealed, tracked instances

transition, under R1/R2 and the permutation gate.

I.2 World models, predictive learning, and consequence forecasting

A long lineage predicts the *latent* state rather than the surface: predictive coding [19,39]; world-model agents that plan over learned latent dynamics [12,37,38], with hierarchical variants planning over abstract latent subgoals [77]; joint-embedding and contrastive self-supervised methods [44,45,49] and their information-theoretic and recurrent-world-model precursors [47,48]; LeCun’s JEPA program [33–36], a *training framework* that predicts a target’s *embedding* (optionally action-conditioned [36,46]) rather than reconstructing pixels or tokens, now extended to language targets [79]; and vision-language-action policies, in which broad, dexterous action emerges from multimodal policy learning [50,51], increasingly coupled to generative subgoal models [78]. The motivation bears on our object: predicting in embedding space keeps salient structure and discards the unpredictable — the same reason TargetSpace scores *transitions in a latent target state* rather than full-future reconstruction.

Reactive, predictive, planning, and hybrid systems are all candidate participants, not competitors, and we take no side on which will prevail. Uniformly, though, they are compared by representation quality, reward, task success, action accuracy, or rollout quality — evaluations that do not, by themselves, test calibration, skill over a target-specific routine, instance specificity, prospective sealing, evidence-value attribution, or multi-horizon target-state forecasting. TargetSpace supplies that measurement layer and applies it to these classes on identical sealed forecast instances; what matters for scoring is whether the predicted transition resolves correctly against the external target, not how the prediction is represented internally.

Table 11: Positioning, not a ranking. The three families are complementary; TargetSpace does not solve world models and takes no position on the eventual architecture. It is architecture-neutral (Section K.1) and asks whether passive observation improves calibrated prediction of human goal-state transitions.

Dimension	Conventional benchmark	LLM	Robotics / world-model benchmark	TargetSpace
Input	a prompt / fixed context		state + action (sim or embodied rollout)	passive observation of a tracked instance up to a sealed time T
Prediction target	next token / answer		next state; consequence of an action (often in latent space)	latent target state and its transition
Evaluation object	output correctness / quality		task success; rollout / planning accuracy	calibrated skill over R1/R2 + permutation specificity
Time horizon	single turn / short		short-to-medium rollouts	hours-weeks; walk-forward, sealed
Role of language	central (it is the task)		peripheral (instructions)	one evidence stream among many; not the success criterion
Role of passive observation	none		sensor stream for control	central — the substrate; its value is measured by the evidence-tier ablation
Failure mode	fluent but wrong; contamination		reconstructs detail; rollout drift; sim-to-real gap	predicts the average not the target (fails R2 / survives permutation)
Success criterion	matches reference / human preference		reaches goal state; low rollout error	calibrated skill over R2 + permutation pass (beats routine, stays calibrated, is target-specific)

I.3 Forecasting, person modeling, and goal recognition

Calibrated event forecasting is established — ForecastBench [28], Prophet Arena [40], and others — but scores external public events, not target-state transitions of a tracked instance, and uses no own-routine or permutation control. Person-modeling benchmarks are the nearest in framing: Park et al. [4,5] replicate individuals’ survey answers (their own-consistency reference is the closest cousin to R2, though used as a ceiling, not a baseline); KnowMe-Bench [26] formalizes person understanding but as *retrospective* comprehension, not prospective forecasting; EgoToM [27] infers a camera-wearer’s goals and future actions from egocentric video, but per clip, by accuracy, without calibration or a persistent target; PersonaMem [7] tracks evolving preferences. Related person-modeling and digital-twin efforts — personalization [3], preference modeling [8], digital twins of survey and behavioural respondents [9,10], a foundation model of human cognition [11], and multimodal theory-of-mind question answering [31] — predict held-out responses or preferences rather than sealed future target-state transitions of a tracked instance. Person-specific multi-horizon forecasting with active/passive evidence comparison already exists in digital phenotyping — we concede this and differentiate on proper scoring, calibration, and the permutation gate. Goal- and plan-recognition [29] infer latent target/goal posteriors under partial observability and define how early a target is identifiable; the field also studies online recognition and changing, hierarchical, or partially observable goals, so we do *not* claim prior work assumes goals are fixed. The narrower distinction is in the evaluation object: conventional goal recognition asks *which* goal, from a candidate set, best explains observed behaviour, whereas TargetSpace prospectively scores how a target’s state, attention, and interaction with the environment produce the *next* target-state transition — emergence, stabilization, competition, displacement, abandonment, resumption, or

opportunity-induced creation (Table 9) — under the R1/R2 and permutation controls. We adopt the resolution-timing idea for the metrics and re-cast it as a calibrated, prospective, instance-tracked measurement; the extension from recognizing a current goal to forecasting target-state formation is subordinate to the target-specificity stack, not a separate theory of goals.

I.4 Cross-scale target-state systems

Beyond persons, the target-state idea recurs — active inference and preferred states [19,32], cell-fate prediction [30], and Levin’s morphogenesis [25] (Section 2.1) — which we cite as lineage and sibling domains, not as competing calibrated benchmarks.

I.5 Intelligence measurement and ARC

Our framing is indebted to Chollet’s account of intelligence as skill-acquisition efficiency and to the ARC-AGI line’s contamination-resistant, private-evaluation design [65,66]; TargetSpace shares the commitments to novelty, freshness, and efficiency (the evidence-tier ablation asks how much evidence a system needs, not only whether it eventually succeeds). ARC evaluates adaptation within a given problem frame, whereas TargetSpace asks the earlier question of inferring the latent state and target from observation when the goal is not framed; both are needed, and neither subsumes the other. Because static, fixed-answer benchmarks saturate and can be gamed once their targets become public [16,17,28], the prospective, future-resolved measurement regime is itself part of the contribution, not merely hygiene: target-specificity is scored against outcomes that do not exist at forecast time.

I.6 Comparing TargetSpace with other benchmark families

The relevant benchmark families are **complementary, not adversarial**: a full picture of a system’s world-modelling ability needs several of them, and TargetSpace should be compared with each only on the axes they share. **Physical-reasoning and intuitive-physics** benchmarks are the right instrument for intuitive physics, object permanence, causality, spatial continuity, and physical-law violation. **Video-generation** benchmarks are right for visual realism, temporal consistency, and plausible scene evolution. **Embodied and robotics** benchmarks [50,51] are right for action utility, policy evaluation, and manipulation or control performance. **Forecasting** benchmarks [28,40] are right for probabilistic future prediction, calibration, and temporal uncertainty. **TargetSpace** is the right instrument for target-specific adaptation, longitudinal memory, goal-transition detection, passive-observation learning, multimodal evidence integration, counterfactual target forecasting, and belief updating over time. Table 12 sets these families on a common template and makes the complementarity explicit, extending the three-way positioning of Table 11 to the broader landscape; the TargetSpace row deliberately concedes the capabilities it does not measure.

Adjacent lines the table does not enumerate. A few neighbouring literatures deserve explicit separation. *Predictive representation learning* — JEPA and its video variant V-JEPA [33–36], and contrastive / joint-embedding methods [44] — learns to predict in a latent space; TargetSpace *evaluates* such models but scores the externally resolved target-state transition, not the quality of the learned representation. *World-model and video-prediction benchmarks* [12,37,38,41,42] score rollout or reconstruction quality; TargetSpace scores a sealed, calibrated forecast about a *persistent* target. *Planning benchmarks* score whether a search or policy reaches a goal state; TargetSpace scores the forecast of the transition, not the plan, and does not infer planning ability from task success (Section 2.5). *Agent and task-completion benchmarks* [2] score what a system *does* when instructed; TargetSpace scores what it can *forecast* about a target from passive observation before

Table 12: Benchmark families and how TargetSpace complements them. The families are complementary, not competing: each is the appropriate instrument for a different question, and TargetSpace is compared with them only on shared axes (cf. Table 11). Bracketed citations point to instances already discussed in Section 6; families named at the category level carry none. The final row states the capabilities TargetSpace does *not* measure, by design.

Benchmark family	Primary evaluation object	Typical input → output	What it under-measures	How TargetSpace complements it
Physical / intuitive physics	physical plausibility, object permanence, causality, spatial continuity	short scenes / clips → plausibility or violation judgment	a persistent target; longitudinal adaptation; calibration over time	adds target-specific dynamics over generic physical law
Video generation / prediction	visual realism, temporal consistency, plausible scene evolution	context frames → generated continuation	target identity; sealed prospective scoring; proper calibration	scores latent target-state transitions, not surface reconstruction
Embodied / robotics [50,51]	action utility, policy evaluation, manipulation or control success	proprioception, sensors, actions → actions / achieved configuration	passive longitudinal inference; calibration; an own-routine baseline	scores passive consequence forecasts, not control
Causal / temporal video QA	causal and counterfactual reasoning over events	video + question → answer (accuracy)	a persistent target; prospective sealing; calibrated baselines	makes causal-style queries prospective, target-specific, proper-scored
Event / symbolic forecasting [28,40]	probabilistic prediction of public events	question + context → calibrated probability	a tracked individual target; own-routine R2; permutation specificity	adds the target as the unit, with R2 and permutation controls
TargetSpace (this work)	target-conditioned longitudinal world modeling	passive multimodal observation up to sealed T → calibrated forecast over target-state transitions	by design: physical realism, control, generation fidelity	is the complementary layer the other families omit

any instruction. *Static LLM benchmarks* [20,21,43] score single-turn correctness on fixed items; TargetSpace is longitudinal, prospective, and sealed. *Longitudinal-memory and personalization benchmarks* [7,26] score recall of, or held-out responses for, an individual; TargetSpace scores whether stored history is converted into improved *future* target-state forecasts under the permutation gate (Section G.2). Finally, *passive-observation datasets* — egocentric video [27], experience sampling [61], personal sensing [62], digital phenotyping [64] — supply the longitudinal signal TargetSpace consumes but are data resources, not a target-specificity evaluation; TargetSpace adds the R1/R2 baselines, sealing, calibration, and permutation test that make target-specificity the measured quantity.

J Extended metrics and scoring details

This appendix carries the metric extensions and reporting details beyond the core scoring of Section 5.

Reporting and inference details

Forecasts use a pre-registered nonzero probability floor; Skill is aggregated within homogeneous answer-space and horizon strata before pooling; Brier is a secondary robustness score, ordered answer spaces may add an ordered proper score, and no composite score is used. R1’s exact fitting

is pre-registered: population information only, leave-one-out with respect to the scored target, rolling, with smoothing for rare classes. Any recalibration uses a separate pre-seal split, is disclosed, and is never fitted on sealed outcomes. R2’s exact feature set, recency kernel, minimum history, and admission rule are pre-registered; the default recipe (marginal frequencies, persistence, recent transitions, time-of-day and day-of-week effects, and recurring calendar/deadline commitments known before the forecast) is Appendix C, organizer-fit rather than entrant-fit, and never selected on test results. The permutation gate swaps complete eligible histories rather than scrambling outcomes and is reported as an effect — true-instance Skill minus the mean permuted-instance Skill, as absolute and proportional skill loss with an interval and a pre-registered margin; its resolving power grows with the number of instances permuted, so it is directional and operational in a five-person pilot and a powered test only above a stated minimum cohort size. Uncertainty respects the repeated, serially dependent structure of forecasts within a target: within-person differences use day-blocked analysis and between-person summaries use person-clustered (cluster-bootstrap) intervals, since the person — not the individual forecast or day — is the independent unit for between-person claims; the current reference implementation reports point estimates.

Multiscale consistency (proposed components)

Because levels are scored separately, we also specify cross-level checks as future evaluation components, not results: *cross-horizon consistency* (a detailed forecast should not contradict the abstract target state without explicit uncertainty or branching); *refinement* (as the horizon shortens and evidence arrives, abstract forecasts should sharpen into concrete ones); *subgoal coherence* (predicted intermediate states should form a plausible path to the higher-level target); *horizon-conditioned calibration* (confidence is evaluated separately at each horizon); *hierarchical abstention* (a system may abstain at the detailed level while keeping a calibrated abstract forecast); and *rollout drift* (repeated one-step predictions may diverge from external reality even when each step is locally accurate [76]).

J.1 Metrics for possibility-space resolution

Predictive lift by evidence tier is Skill computed as a function of the evidence rung — the evidence-ablation read-out; no new mathematics. **Top- k target-state recall** complements distribution-level Skill for large answer spaces. **Transition-path likelihood** is the length-normalized likelihood of the realized path of target-state transitions, in bits vs R1/R2 — the natural metric for the transition track. **Time-to-correct-resolution** and **false-resolution rate** measure *when* a model resolves the possibility space: respectively, how early its distribution concentrates correctly on the realized target, and how often it concentrates confidently and early on the *wrong* one. We are explicit that ‘how early can a target be identified’ is **not** a new idea: it descends directly from goal-recognition design and worst-case distinctiveness (WCD) [29] and from online goal recognition (recognition time, false-positive rate). Our contribution is to operationalize it as a *proper-scored, calibrated, prospective* quantity on tracked instances, paired so that a model must achieve early correct resolution *and* low false resolution together. We freeze and unit-test these definitions before claiming them and report none as results pre-pilot.

K Evaluation grid, evidence ladder, and tracks

This appendix specifies the evaluation grid summarized in Section 3: the architecture classes, the person-domain evidence ladder with its privacy-risk taxonomy, and the multi-track apparatus.

Table 13: Metric provenance. The honesty rule: claim the conjunction and the R2 + permutation anchors; concede the rest as adopted, adapted, or (for resolution-timing) a new operationalization of an existing idea.

Metric	Status	Nearest prior art (cite, do not claim)
Skill / lift / entropy reduction	adopted (information gain)	skill score; Gneiting & Raftery [15]
Log + Brier, ECE gate, sealing	adopted	[14,18,28]; calibration-as-axis [43]
R2 own-routine + permutation gate	anchor (this work)	own-history baselines & self-consistency [5] as cousins
Top- k recall; transition-path likelihood	adapted	retrieval; cell-fate / trajectory likelihoods [30]
Time-to-correct-resolution	operationalization, not new idea	goal-recognition design / WCD [29]; online goal recognition
False-resolution rate	new metric (concept has precedent)	premature-commitment / false-stop / FPR in goal recognition

K.1 Architecture classes

The grid below does not rank architecture classes in the abstract; it has nothing to say about which class is “better” in general. Systems become comparable only because each one emits, or is adapted to emit, a calibrated probability distribution over the *same* answer space A on the *same* sealed instance, scored by the same proper rule. A reactive policy, a large language model, a vision-language model, a JEPA-style latent-predictive model, a symbolic or probabilistic system, a hybrid, and a human filling in a self-report are all evaluated at the level of the forecast output, not by inspecting internal representations.

Because the output adapter can change measured performance, it is disclosed and reported as part of the system under test. An output adapter maps a system’s native output — an action, a token sequence, a latent prediction, a logical query result — to a probability distribution over A . The same underlying model paired with two different adapters is two different entries, and any tuning, temperature setting, or aggregation rule inside the adapter is part of what the score measures. We therefore require the adapter to be specified before the forecast is sealed, so the reported skill in bits over a reference reflects the system as actually run rather than a post hoc reinterpretation of its outputs.

K.2 The evidence ladder

The evidence ladder is the benchmark’s measurement instrument (Figure 5). Tiers are cumulative; we give the canonical *person-domain* ladder and other domains substitute their own sensor ladders (Section 4.3), preserving the ordering from already-interpreted residue toward pre-interpretive observation.

The ladder also encodes a distinction richness alone does not. Lower tiers (L0–L1) are records the target already produced by deciding what to write, send, or schedule — they begin *after* relevance was assigned, and largely encode routine. Higher passive tiers are **pre-interpretive**: captured before the target fixed what would matter, they carry **retrospective option value** — the same recording can be re-scored under questions chosen later. Pre-interpretive capture is not thereby ‘objective’; it is, more precisely, *independent from retrospective human interpretation*. The target-specific hypothesis is sharp: digital exhaust may predict routine; passive evidence may reveal deviations from routine — exactly the transitions where skill over R2 is earned. The grid is the

Table 14: Architecture classes. LLMs and JEPA-style models are complementary components, not rivals: TargetSpace asks whether *any* of them produces calibrated, target-specific forecasts. Human self-report and the oracle anchor the achievable range and are not ranked. The third column indicates what each class supports natively versus through a disclosed output adapter; labels are cautious, classes are not ranked, and real systems are often hybrid.

Architecture class	What it is	Forecast interface (native / via adapter)
LLM-only	text-in, distribution-out language model [20,21,43]	distribution native; explicit state model via adapter
VLM	vision-language model over image / video [31,49]	distribution via adapter; no native state model
Multimodal agent	tool-using agent with retrieval and memory [90,94]	distribution via adapter; target-specific memory possible
JEPA-style / latent predictive (incl. world models)	learns latent dynamics, predicts in representation space [33–36]	state and action-conditioned prediction native; calibrated distribution via adapter; planning by search
Symbolic / probabilistic	explicit structured or Bayesian model [88,95]	calibrated distribution and state prediction native; planning architecture-dependent
Reactive / imitation policy (incl. VLA)	maps observation + instruction to action [50,51]	action native; consequence distribution via adapter; not intrinsic
Hybrid	any combination, e.g. policy + learned subgoal or world model [38,78]	architecture-dependent; reactive and predictive components may coexist
Human self-report (<i>baseline</i>)	the target, or an expert, predicts itself [28,58]	distribution via elicitation; not ranked
Oracle upper bound (<i>baseline</i>)	a privileged predictor; approximate ceiling (benchmark construct, no external referent)	not ranked

cross-product: each leaderboard cell names an (evidence tier, architecture class) pair, with R1, R2, human self-report, and the oracle spanning all tiers.

Departures from an individual’s established routine are not, by default, noise. Research in neuroscience and complex-systems biology indicates that within-system variability can carry information about latent state and cognitive condition [67], and arises from multiple sources that need not be mere nuisance variance — some of it functional, with possible roles in adaptation or exploration [68]. Distinct internal configurations can also produce similar observable outputs, a property known as **degeneracy** [69] — the capacity of structurally different elements or mechanisms to perform the same function or yield the same output (not the model collapse, numerical degeneracy, or redundancy the term denotes in machine learning). In neural systems this is concrete: closely matched network activity can arise from substantially different underlying circuit parameters [70]. (These results concern neural and biological systems; we extend them only by analogy to behavioural forecasting.) Two targets may therefore reach the same state along materially different trajectories, so population averaging or endpoint-only evaluation can erase precisely the person-specific structure that signals an emerging target-state transition. TargetSpace accordingly *preserves* a departure from an individual baseline so the evidence remains available for analysis; this never alters a sealed forecast or its scored target, both fixed at the forecast time (Section 2.3). Whether a departure mattered is asked of preserved evidence in separate, clearly-labelled exploratory analysis, not a

Table 15: The person-domain evidence ladder. Tiers are ordered by increasing capture intrusiveness (and, roughly, privacy sensitivity) — from already-externalized digital residue toward pre-interpretive observation; they are *not* ordered by assumed forecasting power, which is exactly what the ablation measures. By reporting target-specific skill (over R2) as a function of tier, TargetSpace turns ‘does richer observation add target-specific information?’ into a measured quantity rather than an argument.

Tier	Evidence (person domain; operational examples)	Primary sensitivity risks
L0	low-content behavioural metadata: calendar timestamps, communication metadata, app/event logs, coarse routine traces	social-graph and rhythm inference
L1	user-authored text: notes, chats, documents, search/query text, task lists	third-party content; workplace/confidential exposure
L2	speech and transcript layer: audio recordings, ASR transcripts, speaker turns, optional prosody	bystander capture
L3	screen and interaction context: screen recordings, UI activity, browser/app context, document-interaction traces	workplace/confidential exposure; intimate-behaviour inference
L4	egocentric / passive multimodal observation: wearable or scene video, ambient audio, images, environmental context	bystander capture; re-identification
L5	location, mobility, and physical-world traces: GPS, Wi-Fi/Bluetooth proximity, visits, commute patterns	location-trace risk; re-identification
L6	physiological / biometric / specialized sensors: heart rate, sleep, gaze, motion, wearable health signals	health and biometric inference

post-hoc relicensing of the primary score.

The ladder is more than a convenience stratification; it reflects different partial, mediated windows on the target. Textual records are *translated* traces — experience already rendered into language; logs and digital exhaust are *behavioural* traces; audio and video are *temporal observational* traces; and action-conditioned or interventional evidence exposes how the target responds when the world pushes back. No band is reality, and a higher tier is not automatically better — whether it pays is the empirical question the ablation answers. Because the relevance of an observation may only become clear retrospectively, TargetSpace treats raw-evidence preservation, provenance, and ablation as first-class evaluation concerns: the benchmark can ask not only whether a system used evidence but *which* evidence was necessary to improve a target-specific forecast. Preservation is always bounded by the consent, federation, aggregate-only, and retention limits of Section 7; the benchmark scores evidence value, it does not license indiscriminate retention.

Domain tracks

The five domain tracks share the scoring spine (Section 4.3); readiness is stated honestly per track.

L Industry case cluster: from AI recorders to first-person memory systems

This appendix gives the analytical survey behind the motivation of Section 1.1. It is evidence that the capability class is arriving, not empirical validation of TargetSpace, and not a claim that any

Table 16: The five TargetSpace tracks, one shared spine. *Status* — **current**: apparatus exists (synthetic, pre-pilot) for the person track only; **planned**: strong sealed precedent and a natural own-routine baseline, not yet implemented (energy/grid: GEFCom [52]; physiology: PhysioNet/CinC [53], OhioT1DM/BGLP [54]; personal: GLOBEM [55]); **research**: a distinct regime exists but a proper-scored forecasting protocol and/or a strong R2 are not yet established. We make no claim that any track other than the synthetic person track is implemented, and no claim that TargetSpace solves robotics, healthcare, energy, or enterprise forecasting.

Track (<i>status</i>)	Target object	Example evidence bands	Horizon	Example target states	Validator / outcome
TS-Personal (<i>current</i>)	a consenting individual	metadata, text, audio, passive multimodal, location, physiology	hours–weeks	commitment status, priority shift, routine deviation, task switch	observed action / confirmation
TS-Health (<i>planned</i>)	a patient	vitals, labs, continuous glucose monitoring (CGM), wearables, notes	minutes–days	glycemic excursion, deterioration onset, care-state transition	clinical onset labels / sensor thresholds
TS-Energy (<i>planned</i>)	a series / asset	history, weather, calendar, exogenous covariates	hours–days	load / renewable level band, price regime	realized value / market settlement
TS-Robotics (<i>research</i>)	embodied agent + scene	proprioception, sensor stream, action log	sub-second–minutes	goal-conditioned configuration, subgoal transition	achieved configuration
TS-Enterprise (<i>research</i>)	a project / team / workflow	trackers, commits, comms metadata, releases	days–quarters	milestone state, scope / priority drift	observed milestone / outcome

vendor misuses data; each system also has clearly beneficial uses (accessibility, memory support, productivity, care coordination, safety).

Audio-first ambient memory systems

Omi (Based Hardware) is the clearest explicit statement of the industry thesis [96]: its manifesto frames personal context as the competitive moat (“context is the new moat”), describes the product as a memory loop of capture, structure, retrieval, and action rather than a recorder, and names continuity — “memory, goals, patterns” — as the requirement current prompt-response AI lacks. Bee ships a low-cost, continuously listening wristband that converts ambient conversation into summaries, reminders, and to-do items; its acquisition by Amazon in 2025 signals mainstream distribution for the category [97]. Limitless offered a pendant for continuous conversational memory — capture now, search and summarize later — and was acquired by Meta in late 2025 [99]. PLAUD’s press-to-record devices represent the polished episodic end of the spectrum (meetings, calls, notes) [98]; their commercial normalization matters because episodic capture builds the habits, form factors, and supply chains that ambient capture inherits. The relevant TargetSpace observation is common to all four: the product value is increasingly located in the derived, longitudinal layer (summaries, entities, commitments, routines, memory), not in the recording itself — exactly the layering of Table 2.

Audiovisual and egocentric capture

Ray-Ban Meta glasses put camera-and-microphone AI capture into mainstream retail distribution, with a display variant from 2025 [100]. Brilliant Labs’ Halo is an open-source AI-glasses platform whose agent (“Noa”) advertises cross-session narrative memory — an explicit long-term memory claim in a consumer visual device [101]. Snap’s Specs line targets see-through AR with onboard perception [102]. Egocentric-video research (e.g. Ego4D and its successors [56,57]) supplies the datasets, tasks, and methods for exactly this first-person stream. The addition of vision is not merely more data: it adds embodied context — what the wearer sees, where they are, which objects, screens, documents, and people are present, and what actions are physically taken — which is why the personal-track evidence ladder places it at higher tiers (L3–L4) with correspondingly higher sensitivity (Appendix K).

Reading the cluster

Across both groups the trajectory is consistent: from episodic transcription, to persistent personal memory, toward anticipatory assistance — from recording what users say, to modeling what users experience, to predicting target states from longitudinal lived traces. Table 17 contrasts the traditional product framing of each layer with the TargetSpace framing under which it becomes an evaluation and governance object. TargetSpace’s role is to make the top of this ladder measurable: what does longitudinal capture actually buy, over population base rates (R1) and over the target’s own routine (R2), under calibration and permutation specificity? The devices surveyed here are only the most visible edge of the class: the same ladder is climbed, with different signals, by assistants and agents over email, calendar, and documents, by enterprise copilots and organizational memory, by tutors, care systems, and recommenders — any system that converts accumulated longitudinal traces into a persistent, future-relevant model of a particular target (Section 2).

Table 17: From retrieval to experience modeling: the same artifacts read as product features versus as TargetSpace evaluation objects.

Layer	Traditional framing	TargetSpace framing
Audio	recording	behavioural signal
Transcript	searchable text	temporal evidence
Summary	convenience artifact	compressed experience
Entities	names and topics	relationship graph
Tasks	productivity feature	commitment trajectory
Memory	personalization	longitudinal state model
Proactivity	assistant behaviour	target-state intervention
First-person video	visual record	embodied experience trace